



SACS



Material Take-OFF

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Material Takeoff or MTO program can determine the quantity, weight and cost of plate and member materials required to construct the structure as modeled. Weight control reports based on the structural model and loading can be produced. The program can determine the quantity and cost of unmodeled items such as anodes and weld material.

MTO can be coupled with the redesign capability of the SACS system to provide the designer with cost data for various structural configurations during the design process.

1.2 PROGRAM FEATURES

MTO is completely compatible with standard SACS input files and has the following capabilities:

1. Generates list of material by section type or group type.
2. Calculates wetted surface area.
3. Computes weld volumes.
4. Calculates cathodic protection requirements.
5. Performs cost analysis of the structural material, weld material, and the required anodes.
6. Allows for cost overrides for member groups, plate groups, and yield strength.

The weight control feature includes the following capabilities:

1. Creates reports for various structure configurations defined by including or excluding element groups, individual elements, load cases, and disciplines.
2. User defined disciplines, items, and objects.
3. Can access external discipline/item library file.
4. Configuration reports can be sub-divided by discipline and/or item and/or object.
5. Reports weight and C.G. with and without weight contingency factors applied.

2 MTO INPUT DATA

The MTO program requires that program data and options be designated in an MTO input file. The following sections discuss the MTO input data.

2.1 GENERAL PARAMETERS

The Material Take Off program requires specification of some general parameters using the MTOPT line in the MTO input file.

Specify the vertical coordinate in columns 9-10. The water depth and mudline elevation are designated in columns 30-36 and 37-43, respectively. Enter the material density in columns 44-50. The following designates that +Z is the vertical coordinate, the water depth is 150.0, the mudline elevation is -150.0, the material density is 490, and piles are to be included

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	MTOPT	+Z	0	SM	150.0	-150.0	490.0	E

2.2 MEMBER LENGTH OPTION

The actual member length (face to face length) instead of joint to joint length should be used for the purpose of material takeoff and cost estimates. The program has numerous methods available to calculate the member length to use.

One of the following procedures used to calculate the actual member length should be specified in column 13 of the MTOPT line.

1. Specify '0', if a joint thickness specified on the member group line is to be used to calculate the length at each end of the member to be subtracted from the member modeled length.
2. Specify '1' if the joint thickness is to be determined by the MTO program based on the chord diameter. The total joint thickness is taken as the sum of the calculated thickness and the joint thickness specified on the member group line.
3. The joint thickness at each end of the member is determined based on the chord diameter and subtracted from the member modeled length when '2' is specified. Any joint thickness specified on the member group line is ignored.
4. Specify '3' if the actual modeled member length is to be used. Use this option when the model contains the appropriate member offsets or eccentricities.

Note: The term 'modeled member length' refers to the length of the member as it is modeled in the SACS input file and includes the effects of any specified member offsets.

2.3 REPORT LEVEL

Select the output report level, either summary, full or diagnostic by specifying 'SM', 'FL' or 'DG', respectively, in columns 15-16 on the MTOPT line.

2.4 ELEMENT OVERRIDES

Elements groups can be excluded from all calculations by designating the member group on the MGPOVR line, or the plate groups on the PGPOVR line, or the pile group on the PILOVR line by specifying 'X' in column 7. Specific members and plates elements can also be excluded from all calculation by specifying 'X' in column 7 on the MBROVR, and PLAOCR line respectively. For example, member groups PL1 and PL2 representing the piles within the jacket legs are excluded, as well as member group W.B representing a wishbone group.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	MGPOVR	X	PL1	PL2	W.B			

Member groups, Member elements, and Pile groups can also be overridden to include both the external and internal surface area used in anode calculations by specifying 'S' in column 7 of the MGPOVR, MBROVR, and PILOVR line. For example, member 1001-1002 represents a conductor bell guide.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	MBROVR	S	1001	1002				

Plate groups and plate elements can be overridden to include surface area factors by specifying an area factor in column 63-68 of the PGPOVR line for plate groups and PLAOCR line for individual plate elements. For example, plate group PLT has a 2.0 surface area factor since it is exposed to seawater on both sides.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	PGPOVR	PLT					2.0	

2.5 COST OVERRIDE

The material cost specified on the MTOPT line can be overridden for specific elements or groups by using the MGPOVR line for member groups, the PGPOVR line for plate groups, the PILOVR line for pile groups, the MBROVR line for specific members, and the PLAOCR line for specific plates. Also member, plate, pile, and shell groups can have cost overrides based on yield stress with the CSTYLD line. All members, plates, pile, and shells having the same yield stress indicated on the CSTYLD line will have their cost overridden.

2.6 ANODE SPECIFICATIONS

The program can optionally determine the number of anodes required in addition to anode material costs based on NACE SP0173-2007 (formerly RP0176-2003) Appendix D and E, DNV-RP-B401, or by the SACS Anode Method.

Enter the anode calculation method on the ANOPT line column 7-8. Enter "SA" for SACS Anode Method, "ND" for NACE Appendix D / DNV, "NE" for NACE Appendix E, or "NB" for both NACE Appendix D / DNV and NACE Appendix E. The default is the SACS Anode Method.

Enter in the design life in column 10-15 to be used in NACE Appendix D / DNV anode calculation. Enter in the Typical design slope in column 17-22 to be used in NACE Appendix E anode calculation. .

Enter "0" in column 24 for a summary print of the Anode Requirement Report. The summary print will display the number of anodes calculated based on the surface area of all members, plates, shells, and piles. Enter "1" in column 24 for a full print of the Anode Requirement Report. The full print will display the number of anodes calculated based on the surface area of each member group, plate group, shell group, and pile group.

Enter the number of wells in columns 26-27 and the current drain per well in columns 29-35 to account for current load imposed by the well casings below the mudline.

An additional current drain can be specified in column 37-42 to take into account non modeled members that may become electrically connected to the cathodic protection system.

Enter "RD" in column 44-45 to round up the number of anodes calculated in the Anode Summary Report to the nearest whole number. For example if the number of anodes required is 100.1 or 100.9, the summary report will show 101 anodes required, and the total cost will be based on 101 anodes.

The following shows the anode calculation method based on both NACE Appendix D / DNV and NACE Appendix E.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	ANOPT	NB	40.0	44.0	1	5	3.0	RD

2.6.1 SACS Anode Method

In order to use the SACS Anode Method to calculate the number of required anodes, a "SA" needs to be enter in column 7-8 of the ANOPT line or left blank. Cathodic protection data is designated versus water depth. For each water depth specified in columns 12-20, the user should specify the amount of material required per unit surface area to provide protection throughout the life of the structure in columns 21-30. The anode size being used should be specified in columns 31-40 on the first depth entered.

The following designates that at depth 10 through 100, 1000 lb/ft² of anode material is required. The size of the anodes are 725 lb.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	ANODE							
2	ANODE	10.0	1000.	725.				
3	ANODE	100.0	1000.					

Note: The anode size is specified for the first depth only. Also note that depths must be input in order of increasing depth.

2.6.2 Anode Material Cost

If the cost of anode material is to be calculated, enter the material cost per unit weight in columns 41-50 on the ANODE line defining the first water depth.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	ANODE							
2	ANODE	10.0	1000.	725.	100.0			
3	ANODE	100.0	1000.					

Enter "ND" in columns 7-8 of the ANOPT line to use the NACE Appendix D / DNV Anode Method. This method calculates the number of anodes required for three different stages of cathodic protection. The first stage, "Initial", is the number of anodes required to initially polarize the structure. The second stage, "Mean", is the number of anodes required to produce the appropriate number of amps of current over the design life of the structure. The third stage, "Final", is the number of anodes required to produce enough current to maintain protection at the end of the design life. The max number of anodes of the three stages should be the number of anodes used for cathodic protection.

For each level, the initial, mean, and final current density should be enter in column 27-33, 34-40, and 41-47 respectively. The user should specify the water resistivity in column 48-54. The water resistivity is a based on salinity/chlorinity and temperature.

The following designates an anode level for the splash zone (EL. +1 to EL. -15), an anode level from the bottom of splash zone to the mudline (EL. -15 to EL. -100), and an anode level below the mudline (EL. -100 to EL. -250).

	1		2		3		4		5		6		7		8	
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	ANLEV	GR1	-1.0	15.0	0.010	0.050	0.070	0.656	0.10							
2	ANLEV	GR1	15.0	100.0	0.010	0.050	0.070	0.656	1.00							
3	ANLEV	GR1	100.0	250.0	0.002	0.002	0.002	0.656	1.00							

2.6.4 NACE Appendix E Anode Method

Cathodic protection data is designated by levels specified on the ANLEV line. The input on the ANLEV line is similar to NACE Appendix D / DNV except that the initial and final current densities are not required.

Enter "NB" in column 7-8 of the ANOPT line in order to calculate the number of anodes by both NACE Appendix D / DNV and Nace Appendix E Anode Method. The anode cost will be based on the method which requires the most number of anodes.

2.7 ANODE GROUP

Anode information is entered on the AGRUP line. The size of the anode including the length, area, core radius, and weight are entered in column 12-17, 18-23, 24-29, and 30-35 respectively. The energy capability (current capacity) of the anode material is entered in column 36-41.

The net driving potential, which is the difference between the structure's design protective potential and the design closed circuit potential of the anode material, is entered in column 42-47.

Enter the utilization factor in column 48-53. The anode utilization factor is a fraction of the anode material that may be utilized for cathodic protection based on the shape of the anode. Note that utilization factors of 0.90-0.95 for standoff-type anodes, and 0.75 to 0.90 for bracelet anodes are common.

Enter the percentage of length at end of life in column 54-59.

If the anode cost is to be calculated, enter the cost per anode in columns 60-65 on the AGRUP line.

The following defines an example of an anode group.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	AGRUP							
2	AGRUP	GR1	96.0117.90	2.0	725.01100.0	0.25	0.90	0.901000.0

2.8 WEIGHT CONTROL REPORT

The Weight Control Report feature is used to help keep track of the structure weight during different phases of the structure. The feature is used to calculate the weight and CG for the different phase of the structure using only one model.

Each phase is called a 'Configuration' (CONFIG line), and it can include certain members, groups, load conditions, etc. For example during installation, the deck topsides and jacket are shipped to the site on a barge and assembled together in the field. Instead of creating multiple models, a 'Configuration' could be set up to calculate the weight of the deck topsides, and another 'Configuration' for the jacket. Each 'Configuration' can be set up to include certain groups (GRPLST line), load case (LCASE line), Disciplines (DISC line), etc. Weight control reports are generated for each structure 'configuration' specified in the input file.

A 'Discipline' is basically used to group loads together and factor them. Loads might need factoring if the load represents for example, the wet weight of a piece of equipment, and the user wants the dry weight of the equipment. The load could be factored instead of creating a separated load condition. Loads are added to a Discipline as 'Items' (ITEM line) or 'Objects' (OBJECT line). Originally the program was set up for only Items. Since the use of Items were limited and caused confusion, the OBJECT line was created basically to replace the ITEM line.

For an 'Item', each LOAD line can be assigned to only one particular discipline and one 'item' by load label on each LOAD line. This basically means that the user has to have the foresight to set up load labels for MTO weight control reporting during model creation, or go and rename each load labels so that they can be used in the MTO weight control. This problem was solved by adding the OBJECT line to MTO.

'Objects' can be used to specify any load condition or load label to any 'Discipline' and can be factored separately. 'Objects' can also be used to specify a "User Defined Weight". An 'Object' is included in a 'Discipline' using the OBJINC line. This way if multiple 'Disciplines' include the same 'Object', the user does not need to create the same 'Object' for each 'Discipline'.

2.8.1 Weight Control Options

The 'WTCTL' input line must be specified in the MTO input file to invoke the weight control features of the program.

The desired reports are selected in columns 8-13. By default, each configuration report includes the weight control details for each discipline making up that configuration. If details for each item and object making up the discipline are desired in addition, 'PT' should be specified in columns 12 and 13.

Beam and plate elements included in the structural model and load lines with blank load labels are assigned to disciplines 'ST' and 'MI' respectively. These default discipline codes may be overridden in columns 21-22 and 23-24 respectively.

Discipline and item descriptions can be specified in external library files or in the MTO input file. Specify 'XF' in columns 31-32 if the discipline descriptions are to be obtained from an external library file containing only discipline description data. 'XF' should be entered in columns 33-34 if the item descriptions are to be obtained from an external library file containing only item description data. If a library file containing both discipline and item descriptions is to be used, specify 'XF' in columns 35-36 and leave columns 31-34 blank.

2.8.2 Weight Control Disciplines

In general, weight control disciplines are used to group, for reporting purposes, structural elements and/or items and/or objects whose weight and location are accounted for by a 'LOAD' line in the SACS input file.

For items, each 'LOAD' line can be assigned to only one particular discipline and one item. This is done by specifying a discipline code in columns 73 and 74 and an item code in columns 76-80 of the load label. 'LOAD' lines that are not assigned to a discipline by the user (i.e. 'LOAD' line with a blank label in columns 73-80), are assigned to the miscellaneous discipline code specified on the 'WTCTL' input line.

For example, the loading described by the first three input lines below will be assigned to the discipline whose code is 'LA' (specified in columns 73-74), the last 'LOAD' line will be assigned to the miscellaneous discipline code specified on the 'WTCTL' line.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD	601		-5.5		GLOB JOIN	LA-SHACK	
2	LOAD	603		-5.5		GLOB JOIN	LA-SHACK	
3	LOAD	605		-5.5		GLOB JOIN	LA-SHACK	
4	LOAD	773		-5.5		GLOB JOIN	LA-SHACK	

Note: Modeled elements including member and plate elements are assigned to the structural discipline whose code is specified on the 'WTCTL' line.

In order to avoid the load label requirement of items, objects can be used instead. Also when using objects, each 'LOAD' line can be assigned to multiple disciplines and multiple objects. User defined

weights can also be included as an object. The objects included for each discipline are specified using the OBJINC line.

Weight control disciplines are defined using the 'DISC' input line. The two character discipline code and a weight contingency factor to be applied to all elements and/or items and/or objects of the discipline are specified in columns 6-7 and 8-14, respectively. Discipline definitions may be specified in the MTO input file or an external discipline library file.

For the example above, discipline 'LA' is defined by the following discipline definition line:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DISC LA	1.00				LIFT APPURTENANCES		

Note: Discipline library files may also contain item definition lines.

2.8.3 Weight Control Items

Each discipline consist of structural elements and/or one or more items. Item codes are used to designate the item that a particular 'LOAD' line in the SACS input file applies to. For example, the following 'LOAD' lines belong to the item with item code 'SHACK' of the discipline whose code is 'LA'

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD	601		-5.5		GLOB JOIN	LA-SHACK	
2	LOAD	603		-5.5		GLOB JOIN	LA-SHACK	
3	LOAD	605		-5.5		GLOB JOIN	LA-SHACK	

Item definitions are specified using the 'ITEM' input line in the MTO input file or in an external item library file or description library file. For the example above, 'DECK LIFTING EYE SHACKLE' is assigned item code 'SHACK' and is specified in the MTO input or external library file as belonging to the discipline whose code is 'LA' (see below).

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	ITEM SHACK	LA				DECK LIFTING EYE SHACK		

2.8.4 Weight Control Objects

Objects can be used to specify specific loads and/or weight to be included in a specific discipline. The object type is specified in column 8 ('L' for a user defined load, or 'W' for a user defined weight.) For each object, the load case and load label (or weight group id and weight id, if object type is a user defined weight) is entered in column 15-18, and column 20-27 respectively. Each object can be factored by specifying the specific factor in column 29-35

Note: In addition to alphanumeric characters, column 15-18 and column 20-27 recognizes two special characters. A '' refers to any single or set of characters, a '?' refers to any single character.*

The following 'LOAD' lines belong to the object 'HNRL', a 1.05 factor will also be applied:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOADCNCMD							
2	LOAD Z E177E178		-0.0150	-0.0150		GLOB UNIF	HANDRL	
3	LOAD Z E170E171		-0.0150	-0.0150		GLOB UNIF	HANDRL	

By using the '*' in column 15 for the load case, all load cases within the model will be considered. This object will include any LOAD line in the model that has 'HANDRL' as the load label.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	OBJECT L HNDR	HANDRL	1.05	HANDRAIL LOADS				

In order to specify which objects are to be included for each discipline, the OBJINC line must follow the DISC line. For example, the object 'HNDR' will be included in the 'NC' discipline.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DISC NC			NON-CODED STRUCTURAL LOADS				
2	OBJINC HNDR							

2.8.5 Weight Control Configuration

A configuration may consist of the complete structural model or a portion for a particular construction or installation phase (i.e. deck lift configuration, deck in-place configuration, jacket lift configuration etc.). Weight control reports are generated for each configuration specified in the MTO input file.

A configuration includes the specified structural elements and/or load cases contained in the structural model file. For each configuration, the weight control report is categorized into the included disciplines which may be optionally subdivided into items and/or objects.

2.8.5.1 Defining a Configuration

The 'CONFIG' input line is used to define a configuration. The name of the configuration is specified in columns 40-80. The weight contingency factor to be applied to all groups and load cases of this configuration is specified in columns 7-12. Whether to include or exclude listed load cases, member groups, members, plate groups, plates, non-grouped plates, piles, and disciplines is input in columns 14, 16, 17, 18, 19, 20, 21, and 22 respectively. The weight of concentric tubular members will be calculated according to the option specified in column 23. Enter 'B' if both outer and inner tubular and grout annulus will be used in the weight calculation, or 'O' for just the outer tubular, or 'I' for just the inner tubular, or 'N' for both the outer and inner tubular.

2.8.5.2 Configuration Contents

Load cases to be included or excluded in the configuration weight control report are listed on the 'LCASE' input line following the 'CONFIG' line.

Member groups and specific members to be included or excluded in the configuration weight control report are listed on the 'GRPLST' and 'MBRLST' input line following the 'CONFIG' or 'LCASE' line. Plate groups and specific plates that are to be included or excluded in the configuration weight control report are listed on the 'PGRPLST' and PLALST input line. Pile groups that are to be included or excluded in the configuration weight control report are listed on the 'PILLST' input line. Disciplines that are to be included or excluded in the configuration weight control report are listed on the 'DISLST' input line

3 COMMENTARY

3.1 CONNECTION GEOMETRY

The MTO program is designed to calculate plate and member material quantities, surface areas, anode requirements and weld volumes. For members, each end is designated as a brace or chord depending on the relative sizes of all members connecting at that joint. In general, the largest member connecting at the joint is considered to be the chord and all other member ends are designated as braces. Any member the same size as the chord and lying in nearly a straight line with the chord, is considered as a continuation of that chord.

3.1.1 Brace to Chord Connection

MTO computes the geometry of the connection, including the weld length of the brace to the chord, from the SACS model. For a tubular brace, the cross section is divided into points along the center of brace tubular wall. For non-tubular braces, each component of the brace cross section (flange, web, etc.) is divided into points along the component center line. Each point is cast along the brace onto the chord member surface, and the intersection point is found. The weld length is based on the summation of the distance between intersection points.

Figure 1 below shows the weld length between a tubular brace and tubular chord.

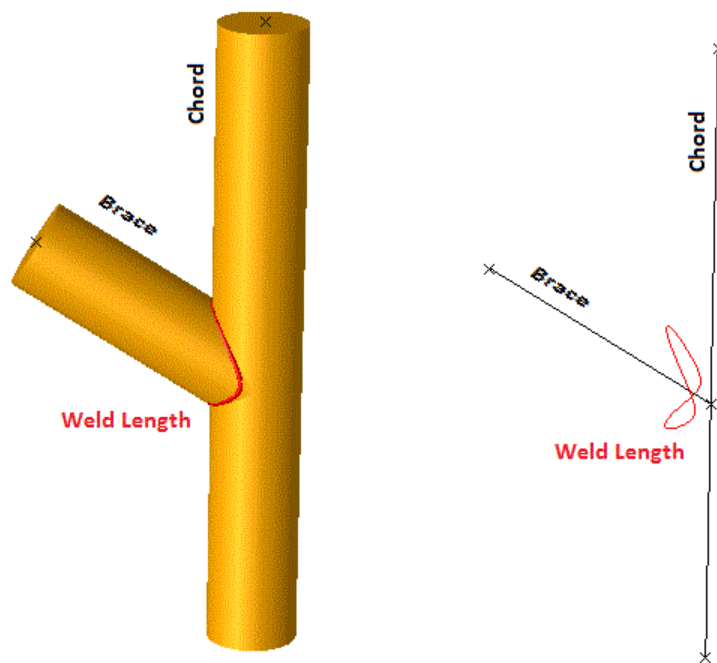
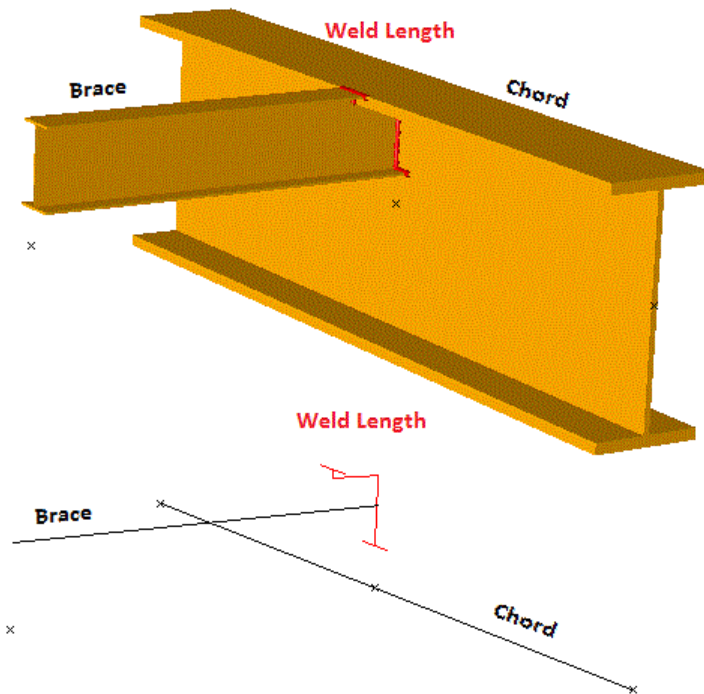
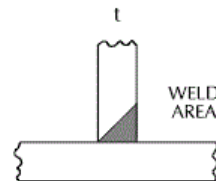


Figure 2 below shows the weld length between a wide flange brace and wide flange chord.



The weld area is computed as a triangular area based on the brace wall or component thickness as follows:

$$A_{weld} = \frac{t_{brace}^2}{2}$$



Weld volume is calculated for the following cross sections: tubular, wide flange, channel, angle, box, prismatic, tee, and double angle.

The weld length for general prismatic members is based as the perimeter of the prismatic footprint on the chord. The weld area is assumed to be one square inch.

4 MISCELLANEOUS

4.1 Welds on Segmented Members

The weld size between two sections of a segmented member is based on the thickness of the thinner cross-section. For example, for a tubular member with a 48" x 1.0 inch segment butt welded to a 48" x 0.5 inch section, the weld volume calculation at the intersection is based in the 0.5 inch wall thickness.

4.2 Plates

Plate areas are calculated along with the equivalent plate thickness (including stiffeners if any) so the weight, surface areas, and material requirements are provided. No attempt is made to calculate plate weld volumes since plate boundaries in an idealized model do not normally follow weld lines.

4.3 Anodes

The surface areas of all elements, except for those excluded on the MGPOVR, PGPOVR, PILOVR, MBROVR, and PLAOVR line, or elements not lying within the specified depth range, are included when determining the number of anodes required for that depth range.

5 ANODE CALCULATION

The MTO program is designed to calculate the required number of anode used to protect submerged steel from corrosion. The surface areas of all members, plates, shells, and piles, except for those excluded, lying with the specified depth range are included when determining the number of anodes required for that depth range.

5.1 SACS Anode Method

In the SACS Anode Method, the user supplies the amount of anode material required per unit surface area for a certain depth range. With the surface area calculated by the program, the number of required anode can be calculated.

$$\# \text{ of Anodes} = \text{Anode Requirement} / \text{Surface area}$$

5.2 NACE Appendix D / DNV Anode Method

In the NACE Appendix D / DNV Anode Method, the number of anodes are calculated for three different stages of cathodic protection: Initial, Mean, and Final. NACE sites one disadvantage of this method is that "it is based on an algorithm rather than being first principles-based and can lead to unnecessary over-design compared to the slope parameter method (See Appendix E)".

Initial

$$\# \text{ of Anodes} = \frac{\text{Initial Current Density} * \text{SA} * \text{Anode Resistance}}{\text{Net Driving Potential} * 1000}$$

$$\begin{aligned} \text{Anode Resistance} &= \frac{\rho}{2 * \pi * L} * \left[\ln \left(4 * \frac{L}{r} \right) - 1 \right] \quad \text{For } L \geq 4r \\ &= 0.315 * \frac{\rho}{A^{.5}} \quad \text{For } L < 4r \end{aligned}$$

Where:

ρ = water resistivity

L = Anode Length

r = Anode radius = $\left(\frac{\text{Anode Area}}{\pi} \right)^{0.5}$

A = Exposed Area = $2 * \pi * r * L$

SA = Surface Area * Surface area factor

Mean

$$\# \text{ of Anodes} = \frac{\text{Mean Current Density} * \text{Life} * \text{SA}}{\text{Energy Capacity} * \text{Weight} * 1000}$$

Where:

$Life = Design\ Life * 8760$

$Weight = Anode\ Weight$

$SA = Surface\ Area * Surface\ area\ factor$

Final

$$\# \text{ of Anodes} = \frac{Final\ Current\ Density * SA * Anode\ Resistance}{Net\ Driving\ Potential * 1000}$$

$$Anode\ Resistance = \frac{\rho}{2 * \pi * L} * \left[\ln \left(4 * \frac{L}{r} \right) - 1 \right] \quad \text{For } L \geq 4r$$
$$= 0.315 * \frac{\rho}{(A^{.5})} \quad \text{For } L < 4r$$

Where:

$\rho =$ water resistivity

$L =$ Anode Length * % of Length at End of Life

$r =$ Expanded Anode Radius

$r =$ Anode radius - (Anode radius - Anode Core Radius) * Utilization Factor

$$Anode\ radius = \left(\frac{Anode\ Area}{\pi} \right)^{0.5}$$

$A =$ Exposed Area = $2 * \pi * r * L$

$SA =$ Surface area * Surface area factor

Note: In order to calculate the number of anodes required for the additional Current Drain for each stage, replace the (current density * SA) with the additional current drain specified on the ANOPT line. The anode properties used are from the anode group of the last anode level.

5.3 NACE Appendix E Anode Method

In the NACE Appendix E Anode Method, the number of anodes are based on the Design Slope and the Mean current density.

$$\# \text{ of Anodes} = \frac{SA * Anode\ Resistance}{Design\ Slope}$$

$$Anode\ Resistance = \frac{\rho}{2 * \pi * L} * \left[\ln \left(4 * \frac{L}{r} \right) - 1 \right] \quad \text{For } L \geq 4r$$
$$= 0.315 * \frac{\rho}{A^{.5}} \quad \text{For } L < 4r$$

$$Design\ Life\ of\ Anode = \frac{Anode\ Resistance * Weight * 1000 * Energy\ Capability}{Mean\ Current\ Density * 8760 * Design\ Slope}$$

Where:

ρ = water resistivity

L = Anode Length

r = Anode radius = $\left(\frac{\text{AnodeArea}}{\pi}\right)^{0.5}$

A = Exposed Area = $2 * \pi * r * L$

SA = Surface Area * Surface area factor

$Weight$ = Anode Weight * Utilization Factor

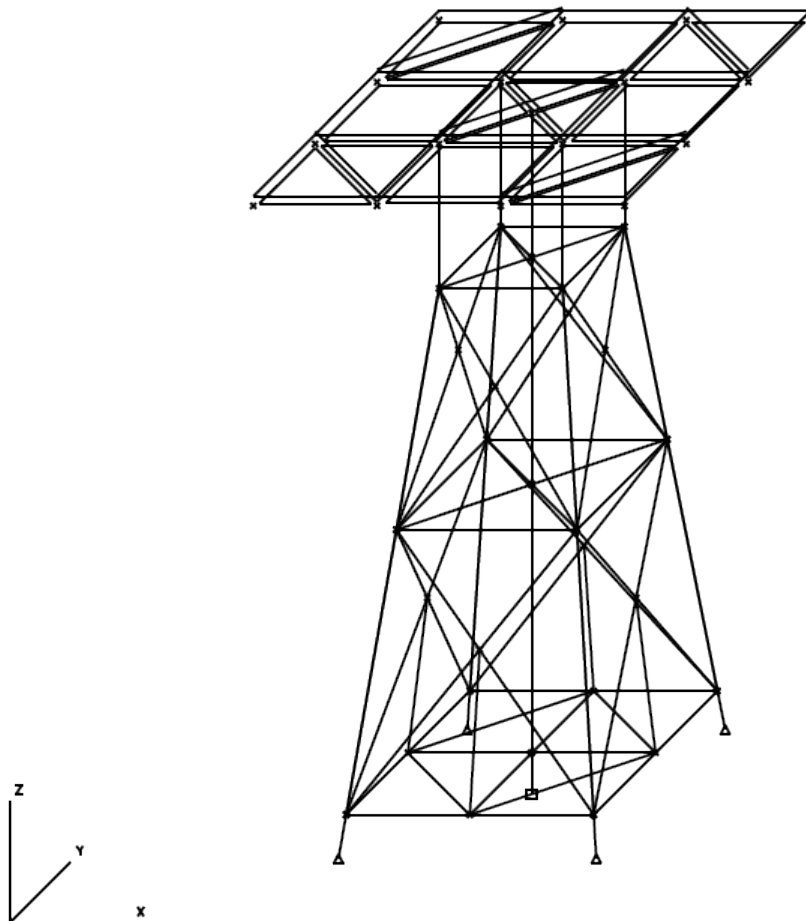
In order to calculate the number of anodes required for the additional Current Drain, the following formula is used. The anode properties used are from the anode group of the last anode level.

$$\# \text{ of Anodes} = \frac{\text{Additional Current Drain} * \text{Life}}{\text{Energy Capacity} * \text{Weight} * 1000}$$

6 SAMPLE PROBLEMS

The structure shown below was used to illustrate some of the features of the MTO program module. The sample structure contains tubular, wide flange and plate elements.

SAMPLE JACKET STRUCTURE



6.1 COST ANALYSIS

The structure stands in 261.0 feet of water and is modeled such that the water surface is at elevation 0.0. The construction cost of tubular steel is assumed to be \$0.95/LB except that the cost of pile steel is \$0.85/LB. The construction cost of wide flange steel is \$1.80/LB. The cost of weld material is assumed to be included in the construction cost of the steel and anode material is assumed to cost \$0.40/LB.

Below is the MTO input file used for this sample problem followed by an explanation of each input line.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	MTOPT	+Z	0	FL	FL	CS	SA	261. -261. 0.95 E
2	ANODE							
3	ANODE		40.0	2.1	725.0	0.40		
4	ANODE		260.	1.2				
5	GRPCST	PL1	160.	PL2	160.	PL3	160.	PL4 160.
6	GRPCST	W01	300.	W02	240.			
7	END							

Line 1. The MTO options line specifies the following:

- The input joint thickness specified on the member group line is to be used to calculate the length at each end of the member as specified by '0' in column 13.
- The full reports are desired ('FL' in col. 15-16).
- Weld volume, cost analysis and surface area reports are requested by 'FL', 'CS' and 'SA' in columns 18-19, 21-22 and 24-25, respectively.
- The water depth is 261. feet and the mudline is at -261. ft.
- The default cost of steel is \$0.95/LB as specified by '0.95' in cols. 51-54.
- Pile elements will be excluded as specified by 'E' in column 72.

Line 2&3. The first ANODE line is a header line. The second specifies the following:

- From 0 to 40 ft. water depth, 2.1 LBS of anode material is required per square foot of surface area.
- 725 LB anodes are to be used as specified by '725.0' in columns 31-35.
- The cost of anode material is \$0.40/LB as specified in columns 41-44.

Line 4. The third ANODE line specifies:

- 1.2 LB/ft² of anode material is required from 40 to 260 foot water depth.

Line 5. The first GRPCST line designates the material price for groups PL1, PL2, PL3 and PL4 shall be \$160/ft.

E. The next GRPCST line specifies that the cost of steel for wide flange groups W01 and W02 shall be \$300/ft and \$240lb/ft respectively.

The following pages contain a portion of the MTO analysis output.

MTO VERSION 14.3.0.27

*** ANALYSIS SELECTIONS

MATERIAL AND WEIGHT TAKE-OFF

SURFACE AREAS

WELD VOLUMES

COST ANALYSIS

ANODE REQUIREMENTS

*** CASE PARAMETERS

NUMBER OF MEMBERS 149

NUMBER OF PLATES 2

NUMBER OF SHELLS 0

NUMBER OF PILES 0

NUMBER OF JOINTS 83

MATERIAL DENSITY 490.00 LB/FT3

CONCRETE DENSITY 145.00 LB/FT3

WATER DEPTH 261.00 FT

MUDLINE ELEVATION -261.00 FT

VERTICAL COORDINATE+Z

MATERIAL COST

BASIC 0.950 \$/LB

TUBULAR 0.950 \$/LB

INTERNAL RINGS 0.950 \$/LB

EXTERNAL RINGS 0.950 \$/LB

CONCRETE 1.000 \$/LB

ANODE MATERIAL 0.400 \$/LB

ANODE SIZE 725.00 LB
 ANODE PLATE AREA FACTOR . 1.00
 WELD COST 0.000 \$/IN**3
 UNITSENGLISH
 PRINT LEVELFULL
 MEMBER ENDUSE INPUT JOINT THICKNESS

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* * * G R U P S U M M A R Y * * *

GRUP CODE	SECTION ID	TYPE	***** A IN	SECTION B IN	DIMENSIONS C IN	***** D IN	E IN	CONST. AREA IN**2	VAR. SECT. FT	JOINT THICK FT	YIELD STRESS KSI	RUNNING COST \$/FT	* TOTAL OUT-OUT FT	LENGTHS * CENTERLN FT	TOTAL COST \$
LG1		TUB	42.00	1.38	0.00	0.00	0.00	175.49	5.00	0.00	50.00	567.29	20.00	20.00	11346.43
LG1		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	351.98	351.98	143876.70
LG1		TUB	42.00	1.38	0.00	0.00	0.00	175.49	5.00	0.00	50.00	567.29	20.00	20.00	11346.43
LG2		TUB	42.00	1.38	0.00	0.00	0.00	175.49	6.15	0.00	50.00	567.29	24.60	24.60	13955.96
LG2		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	339.70	339.70	138856.25
LG2		TUB	42.00	1.38	0.00	0.00	0.00	175.49	4.90	0.00	50.00	567.29	19.60	19.60	11118.83
LG3		TUB	42.00	1.38	0.00	0.00	0.00	175.49	6.75	0.00	50.00	567.29	27.00	27.00	15317.44
LG3		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	260.70	260.70	106563.75
LG3		TUB	42.00	1.38	0.00	0.00	0.00	175.49	4.35	0.00	50.00	567.29	17.40	17.40	9871.48
LG4		TUB	42.00	1.38	0.00	0.00	0.00	175.49	0.00	0.00	50.00	567.29	14.15	14.14	8024.99
LG5		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	50.00	355.45	12.13	12.12	4309.85
LG6		TUB	36.00	0.75	0.00	0.00	0.00	83.06	3.25	0.00	36.00	268.49	13.85	13.00	3717.59

LG6	CONE	CON	36.00	0.75	26.00	0.00	0.00	71.27	4.95	0.00	36.00	230.41	19.80	19.80	4562.04
LG6		TUB	26.00	0.75	0.00	0.00	0.00	59.49	0.00	0.00	36.00	192.32	115.20	115.20	22155.53
LG7		TUB	26.00	0.75	0.00	0.00	0.00	59.49	0.00	0.00	36.00	192.32	100.00	100.00	19232.23
PL1		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	391.98	391.98	62717.32
PL2		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	383.90	383.90	61424.44
PL3		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	305.10	305.10	48816.63
PL4		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	14.15	14.14	2263.79
T01		TUB	16.00	0.62	0.00	0.00	0.00	30.19	0.00	0.00	35.00	97.59	948.82	931.58	92594.34

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* * * G R U P S U M M A R Y * * *															
GRUP CODE	SECTION ID	TYPE	***** A IN	SECTION B IN	DIMENSIONS C IN	***** D IN	***** E IN	CONST. AREA IN**2	VAR. SECT. FT	JOINT THICK FT	YIELD STRESS KSI	RUNNING COST \$/FT	* TOTAL OUT-OUT FT	LENGTHS * CENTERLN FT	TOTAL COST \$
T02		TUB	20.00	0.75	0.00	0.00	0.00	45.36	0.00	0.00	35.00	146.62	753.11	740.94	110422.80
T03		TUB	12.75	0.50	0.00	0.00	0.00	19.24	0.00	0.00	35.00	62.20	284.96	282.83	17725.72
T04		TUB	24.00	0.75	0.00	0.00	0.00	54.78	0.00	0.00	36.00	177.09	254.63	251.28	45092.30
T05		TUB	26.00	1.00	0.00	0.00	0.00	78.54	0.00	0.00	36.00	253.89	869.69	855.83	220806.64
W01	W24X162	WF	12.95	1.22	25.00	0.70	0.50	47.70	0.00	0.00	36.00	300.00	236.44	226.02	70931.59
W02	W24X131	WF	12.85	0.96	24.48	0.61	0.50	38.50	0.00	0.00	36.00	240.00	499.48	472.45	119874.83
SACS CONNECT Edition V(14.3) - CL								Company: Bentley Sytems							
***** SACS IV MTO PROGRAM *****								DATE 31-AUG-2020 TIME 11:13:36 MTO PAGE 10							

* * * G R U P S U M M A R Y * * *

GRUP CODE	SECTION ID	TYPE	***** A IN	SECTION B IN	DIMENSIONS C IN	***** D IN	***** E IN	CONST. AREA IN**2	VAR. SECT. FT	JOINT THICK FT	YIELD STRESS KSI	RUNNING COST \$/FT	* TOTAL OUT-OUT KIPS	WEIGHTS * CENTERLN KIPS	TOTAL COST \$
LG1		TUB	42.00	1.38	0.00	0.00	0.00	175.49	5.00	0.00	50.00	567.29	11.94	11.94	11346.43
LG1		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	151.45	151.45	143876.70
LG1		TUB	42.00	1.38	0.00	0.00	0.00	175.49	5.00	0.00	50.00	567.29	11.94	11.94	11346.43
LG2		TUB	42.00	1.38	0.00	0.00	0.00	175.49	6.15	0.00	50.00	567.29	14.69	14.69	13955.96
LG2		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	146.16	146.16	138856.25
LG2		TUB	42.00	1.38	0.00	0.00	0.00	175.49	4.90	0.00	50.00	567.29	11.70	11.70	11118.83
LG3		TUB	42.00	1.38	0.00	0.00	0.00	175.49	6.75	0.00	50.00	567.29	16.12	16.12	15317.44
LG3		TUB	41.25	1.00	0.00	0.00	0.00	126.45	0.00	0.00	36.00	408.76	112.17	112.17	106563.75
LG3		TUB	42.00	1.38	0.00	0.00	0.00	175.49	4.35	0.00	50.00	567.29	10.39	10.39	9871.48
LG4		TUB	42.00	1.38	0.00	0.00	0.00	175.49	0.00	0.00	50.00	567.29	8.45	8.45	8024.99
LG5		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	50.00	355.45	4.54	4.54	4309.85
LG6		TUB	36.00	0.75	0.00	0.00	0.00	83.06	3.25	0.00	36.00	268.49	3.91	3.67	3717.59
LG6	CONE	CON	36.00	0.75	26.00	0.00	0.00	71.27	4.95	0.00	36.00	230.41	4.80	4.80	4562.04
LG6		TUB	26.00	0.75	0.00	0.00	0.00	59.49	0.00	0.00	36.00	192.32	23.32	23.32	22155.53
LG7		TUB	26.00	0.75	0.00	0.00	0.00	59.49	0.00	0.00	36.00	192.32	20.24	20.24	19232.23
PL1		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	146.66	146.66	62717.32
PL2		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	143.64	143.64	61424.44
PL3		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	114.16	114.15	48816.63
PL4		TUB	36.00	1.00	0.00	0.00	0.00	109.96	0.00	0.00	36.00	160.00	5.29	5.29	2263.79
T01		TUB	16.00	0.62	0.00	0.00	0.00	30.19	0.00	0.00	35.00	97.59	97.47	95.70	92594.34

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* * * G R U P S U M M A R Y * * *

GRUP CODE	SECTION ID	TYPE	***** A IN	SECTION B IN	DIMENSIONS C IN	***** D IN	***** E IN	CONST. AREA IN**2	VAR. SECT. FT	JOINT THICK FT	YIELD STRESS KSI	RUNNING COST \$/FT	* TOTAL OUT-OUT KIPS	WEIGHTS * CENTERLN KIPS	TOTAL COST \$
T02		TUB	20.00	0.75	0.00	0.00	0.00	45.36	0.00	0.00	35.00	146.62	116.23	114.36	110422.80
T03		TUB	12.75	0.50	0.00	0.00	0.00	19.24	0.00	0.00	35.00	62.20	18.66	18.52	17725.72
T04		TUB	24.00	0.75	0.00	0.00	0.00	54.78	0.00	0.00	36.00	177.09	47.47	46.84	45092.30
T05		TUB	26.00	1.00	0.00	0.00	0.00	78.54	0.00	0.00	36.00	253.89	232.43	228.72	220806.64
W01	W24X162	WF	12.95	1.22	25.00	0.70	0.50	47.70	0.00	0.00	36.00	300.00	38.38	36.69	70931.59
W02	W24X131	WF	12.85	0.96	24.48	0.61	0.50	38.50	0.00	0.00	36.00	240.00	65.44	61.89	119874.83

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* * * A N O D E R E Q U I R E M E N T S * * *
(SACS ANODE METHOD)

WATER DEPTH (FT)	STRUCTURAL ELEVATION (FT)	ANODE REQUIRED (LB/SQ FT)	SURFACE AREA (SQ FT)	TOTAL ANODE REQUIREMENT (LB)	NUMBER OF ANODES
0.00	0.00	2.100			
			4065.88	8538.3	11.8
40.00	-40.00	2.100			
			28261.39	33913.7	46.8
260.00	-260.00	1.200			

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***** ANODE SUMMARY *****

TOTAL ANODE MATERIAL 42452.01 LB

TOTAL NUMBER OF ANODES ... 58.55

TOTAL ANODE COST 16980.80

6.2 WEIGHT CONTROL REPORT

Weight control reports are desired for the same structure used in Sample Problem 1 for deck lift, deck in-place, jacket lift and general pile configurations. Discipline codes and item codes are defined together in an external library file.

The 'LOAD' lines in the SACS model file were modified so that the labels in columns 73-80 designated to which discipline and discipline item the 'LOAD' line applied. The loading portion of the model file is below:

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
LOAD								
LOADCN	1							
** DECK UNMODELED DECK BEAMS								
LOAD Z	407 401		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	401 462		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	405 403		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	403 463		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	468 405		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	469 407		-1.476	-1.476		GLOB UNIF	MS-DBEAM	
LOAD Z	472 461		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOAD Z	465 464		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOAD Z	466 465		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOAD Z	467 466		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOAD Z	470 471		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOAD Z	471 472		-0.738	-0.738		GLOB UNIF	MS-DBEAM	
LOADCN	2							
** DECK MISCELLANEOUS ITEMS								
LOAD Z	407 401		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	401 462		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	405 403		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	403 463		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	468 405		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	469 407		-0.492	-0.492		GLOB UNIF	EL-ELECT	
LOAD Z	472 461		-0.075	-0.075		GLOB UNIF	MS-WALK	
LOAD Z	471 472		-0.075	-0.075		GLOB UNIF	MS-WALK	
LOAD Z	470 471		-0.075	-0.075		GLOB UNIF	MS-WALK	
LOAD Z	472 461		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	471 472		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	470 471		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	469 470		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	468 469		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	467 468		-0.015	-0.015		GLOB UNIF	MS-HRAIL	
LOAD Z	465 466	2.00000-3.0000				GLOB CONC	MS-STAIR	
LOAD Z	465 466	6.00000-3.0000				GLOB CONC	MS-STAIR	
LOAD	401		-0.5000			GLOB JOIN	LA-LEYE	
LOAD	407		-0.5000			GLOB JOIN	LA-LEYE	
LOAD	405		-0.5000			GLOB JOIN	LA-LEYE	
LOAD	403		-0.5000			GLOB JOIN	LA-LEYE	
LOAD	461		-10.000			GLOB JOIN	CR-CRANE	
LOAD	472		-2.0000			GLOB JOIN	CR-BREST	

1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890							
LOADCN	3						
** JACKET MISCELLANEOUS ITEMS							
LOAD Z	303	305	2.00000-3.0000			GLOB CONC	MS-STAIR
LOAD Z	303	305	6.00000-3.0000			GLOB CONC	MS-STAIR
LOAD Z	303	305	-0.075	-0.075		GLOB UNIF	MS-WALK
LOAD Z	301	303	-0.075	-0.075		GLOB UNIF	MS-WALK
LOAD Z	303	305	-0.015	-0.015		GLOB UNIF	MS-HRAIL
LOAD Z	301	303	-0.015	-0.015		GLOB UNIF	MS-HRAIL
LOAD	301		-0.5000			GLOB JOIN	LA-LEYE
LOAD	307		-0.5000			GLOB JOIN	LA-LEYE
LOAD	305		-0.5000			GLOB JOIN	LA-LEYE
LOAD	303		-0.5000			GLOB JOIN	LA-LEYE
LOAD	101		-3.0000			GLOB JOIN	MM-MMAT
LOAD	103		-3.0000			GLOB JOIN	MM-MMAT
LOAD	105		-3.0000			GLOB JOIN	MM-MMAT
LOAD	107		-3.0000			GLOB JOIN	MM-MMAT
LOADCN	4						
** DECK EQUIPMENT DRY WEIGHT							
LOAD Z	405	468	11.4075-15.364			GLOB CONC	EQ-BLDG1
LOAD Z	405	468	18.9075-16.826			GLOB CONC	EQ-BLDG1
LOAD Z	466	467	11.4075-26.921			GLOB CONC	EQ-BLDG1
LOAD Z	466	467	18.9075-28.383			GLOB CONC	EQ-BLDG1
LOAD Z	466	468	16.1326-20.224			GLOB CONC	EQ-BLDG1
LOAD Z	466	468	26.7392-17.283			GLOB CONC	EQ-BLDG1
LOAD Z	403	405	9.84250-20.876			GLOB CONC	EQ-SKID1
LOAD Z	405	468	.157500-15.333			GLOB CONC	EQ-SKID1
LOAD Z	465	466	9.84250-16.614			GLOB CONC	EQ-SKID1
LOAD Z	466	467	.157500-11.071			GLOB CONC	EQ-SKID1
LOAD Z	466	468	.222738-11.106			GLOB CONC	EQ-SKID1
LOAD Z	462	401	9.52758-12.586			GLOB CONC	EQ-SKID2
LOAD Z	462	401	19.5276-13.184			GLOB CONC	EQ-SKID2
LOAD Z	472	461	10.1575-9.0445			GLOB CONC	EQ-SKID2
LOAD Z	472	461	.157500-9.6428			GLOB CONC	EQ-SKID2
LOAD Z	472	462	14.3649-10.872			GLOB CONC	EQ-SKID2
LOAD Z	472	462	.222738-9.6712			GLOB CONC	EQ-SKID2
LOADCN	5						
** DECK EQUIPMENT FLUID WEIGHT							
LOAD Z	405	468	11.4075-1.5364			GLOB CONC	EQ-BLDG1
LOAD Z	405	468	18.9075-1.6826			GLOB CONC	EQ-BLDG1
LOAD Z	466	467	11.4075-2.6921			GLOB CONC	EQ-BLDG1
LOAD Z	466	467	18.9075-2.8383			GLOB CONC	EQ-BLDG1
LOAD Z	466	468	16.1326-2.0224			GLOB CONC	EQ-BLDG1
LOAD Z	466	468	26.7392-1.7283			GLOB CONC	EQ-BLDG1
LOAD Z	403	405	9.84250-2.0876			GLOB CONC	EQ-SKID1
LOAD Z	405	468	.157500-1.5333			GLOB CONC	EQ-SKID1
LOAD Z	465	466	9.84250-1.6614			GLOB CONC	EQ-SKID1
LOAD Z	466	467	.157500-1.1071			GLOB CONC	EQ-SKID1
LOAD Z	466	468	.222738-1.1106			GLOB CONC	EQ-SKID1
LOAD Z	462	401	9.52758-1.2586			GLOB CONC	EQ-SKID2
LOAD Z	462	401	19.5276-1.3184			GLOB CONC	EQ-SKID2
LOAD Z	472	461	10.1575-0.9645			GLOB CONC	EQ-SKID2
LOAD Z	472	461	.157500-0.9648			GLOB CONC	EQ-SKID2
LOAD Z	472	462	14.3649-1.0872			GLOB CONC	EQ-SKID2
LOAD Z	472	462	.222738-0.9672			GLOB CONC	EQ-SKID2
LOADCN	6						
** DECK 100 PSF LIVE LOAD							
LOAD Z	401	407	-1.969	-1.969		GLOB UNIF	A100PSF
LOAD Z	462	401	-1.969	-1.969		GLOB UNIF	A100PSF
LOAD Z	403	405	-1.969	-1.969		GLOB UNIF	A100PSF
LOAD Z	463	403	-1.969	-1.969		GLOB UNIF	A100PSF
END							

The following external library file containing the discipline and item descriptions was used:

1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890							
DISC AP 1.0					APPURTENANCES		
ITEM RISE AP					UNMODELED RISERS AND CLAMPS		
ITEM COND AP					UNMODELED CONDUCTORS		
ITEM BOAT AP					UNMODELED BOAT LANDINGS		
ITEM BUMP AP					UNMODELED BARGE BUMPERS		
ITEM MISC AP					MISCELLANEOUS APPURTENANCES		
DISC CR 1.0					CRANE		
ITEM BOOM CR					CRANE BOOM		
ITEM BREST CR					CRANE BOOM REST		
ITEM CRANE CR					CRANE, AND CRANE PEDESTAL		
DISC EL 1.0					ELECTRICAL		
ITEM ELECT EL					MISCELLANEOUS ELECTRICAL		
DISC EQ 1.0					EQUIPMENT		
ITEM BLDG1 EQ					CONTROL & ENGINE BUILDING		
ITEM SKID1 EQ					COMPRESSOR SKID		
ITEM SKID2 EQ					FIRE EQUIPMENT SKID		
ITEM MISC EQ					MISCELLANEOUS EQUIPMENT		
DISC HD 1.0					HELIDECK		
ITEM STAIR HD					HELIDECK STAIRS		
ITEM SNET HD					HELIDECK SAFETY NET		
ITEM WALK HD					HELIDECK WALKWAY		
ITEM MISC HD					HELIDECK MISCELLANEOUS		
DISC LA 1.0					LIFT APPURTENANCES		
ITEM LEYE LA					LIFTING EYES		
ITEM SHACK LA					LIFTING SHACKLES		
ITEM SLING LA					LIFTING SLINGS		
ITEM MISC LA					MISCELLANEOUS LIFT APPURTENANCES		
DISC MM 1.0					MUD MATS		
ITEM MMAT MM					MUD MAT PLATE		
ITEM MSTIF MM					MUD MAT STIFFENERS		
DISC MS 1.0					MISCELLANEOUS STRUCTURAL STEEL		
ITEM DBEAM MS					UNMODELED DECK BEAMS		
ITEM GRATE MS					GRATING		
ITEM HRAIL MS					HAND RAILING		
ITEM PLATE MS					UNMODELED PLATE		
ITEM STAIR MS					STAIRS		
ITEM WALK MS					WALKWAYS		
ITEM MISC MS					MISCELLANEOUS UNMODELED ITEMS		

The 'DISC' line defines the discipline specified in columns 40-80, including the discipline code and the weight contingency factor. The first 'DISC' line specifies the following:

- A. The discipline code for the discipline 'Appurtenances' is 'AP' as designated in columns 6 and 7.
- B. The weight contingency factor to be applied to all items of this discipline is 1.0 as specified in columns 8-14.

The 'ITEM' line defines items belonging to a particular discipline. The item name is specified in columns 40-80 along with the item code and the discipline to which it belongs. The first 'ITEM' line following the 'Appurtenances' discipline line species the following:

- A. The 'Unmodeled risers and clamps' item is assigned an item code 'RISE' as specified in columns 6-10.

B. 'Unmodeled risers and clamps' are designated as part of the 'Appurtenances' discipline by specifying the discipline code 'AP' in columns 14-15.

Note: 'DISC' and 'ITEM' input lines may be specified in any order in the external library file. This allows the user the freedom to assemble data in the most logical sequence for their application.

Weight control input lines were added to the input file used in Sample Problem 1. Below is the input file followed by a detailed description of the input lines applicable to the weight control portion of MTO.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
	MTOPT	3	SM	WV	CS	SA	82.02	-82.02
	ANODE						0.95	0.0
	ANODE	40.0		2.10		725.0	0.40	
	ANODE	80.0		1.20				
	GRPCST	PL1	0.85	PL2	0.85	PL3	0.85	PL4
	GRPCST	DK1	1.80	DK2	1.80			
A	WTCTL	PTPTPT					XF	
B	CONFIG	1.05	I	I	X			DECK LIFT CONFIGURATION
C	GRPLST	DK1	DK2	PL3				
D	LCASE	1	2	4				
E	CONFIG	1.05	I	I	X			DECK IN-PLACE CONFIGURATION
	GRPLST	DK1	DK2	PL3				
	LCASE	1	2	4	5			
F	CONFIG	1.05	I	X	X			JACKET LIFT CONFIGURATION
G	GRPLST	DK1	DK2	PL1	PL2	PL3		
H	PGRPLSTAAA							
I	LCASE	3						
J	CONFIG	1.05	I	I				PILE CONFIGURATION
	GRPLST	PL1	PL2					

A. The WTCTL options line specifies the following:

- Discipline and item descriptions are to be printed in the MTO listing file as specified by 'PT' in columns 8-9 and 10-11 respectively.
- The weight control report for each configuration should be reported by discipline, with details of each item of the discipline ('PT' columns 12-13).
- Model elements will be assigned discipline code 'ST', all 'LOAD' lines without discipline and/or item codes specified in columns 73-80 will be assigned discipline code 'MI' by default.
- The discipline and item definitions are in the same external library file ('XF' columns 35-36).

B. The first 'CONFIG' line defines the 'DECK LIFT CONFIGURATION' as designated in columns 40-80 and specifies the following:

- A 1.05 contingency factor will be applied to all load case disciplines and member and plate elements of this configuration as specified in columns 7-12.

-
- b. 'I' in column 14 designates that discipline and item weights defined by 'LOAD' lines in load cases listed on the following 'LCASE' line are to be included in this configuration.
 - c. 'I' in column 16 designates that only elements in member groups listed on the following 'GRPLST' line are to be considered as part of this configuration.
 - d. 'X' in column 18 specifies that any plate groups listed on 'PGRPLST' lines are to be excluded from this configuration. Plate elements belonging to plate groups not listed are to be included.
- C. The 'GRPLST' line specifies that members belonging to groups 'DK1', 'DK2' and 'PL3' are to be included.
- D. The 'LCASE' line designates that load cases 1, 2, and 4 are to be included in the 'DECK LIFT CONFIGURATION' report.
- E. The second 'CONFIG' line defines the 'DECK IN-PLACE CONFIGURATION' as designated in columns 40-80. This configuration is identical to the 'DECK LIFT CONFIGURATION' except that load case 5 is included in addition to load cases 1, 2 and 4 as designated on the 'LCASE' line.
- F. The 'JACKET LIFT CONFIGURATION' is defined by the third 'CONFIG' line as follows:
- a. 'I' in column 14 designates that discipline and item weights defined by 'LOAD' lines in load cases listed on the following 'LCASE' line are to be included in this configuration.
 - b. 'X' in column 16 designates that elements in member groups listed on the following 'GRPLST' line are to be excluded from this configuration. All elements belonging to groups not listed are to be included.
 - c. 'X' in column 18 specifies that any plate groups listed on 'PGRPLST' lines are to be excluded from this configuration. Plate elements belonging to plate groups not listed are to be included.
- G. The 'GRPLST' line specifies that member groups 'DK1', 'DK2', 'PL1', 'PL2' and 'PL3' are to be excluded.
- H. The 'PGRPLST' line specifies that plates belonging to group 'AAA' are to be excluded.
- I. The 'LCASE' line designates that only load case 3 is to be included in the 'JACKET LIFT CONFIGURATION' report.
- J. The 'PILE CONFIGURATION' is defined such that only groups 'PL1' and 'PL2' are included.

The following pages contain a portion of the MTO weight control output.

*** DISCIPLINE DESCRIPTIONS ***

NO.	CODE	WEIGHT FACTOR	***** DESCRIPTION *****
1	AP	1.0000	APPURTENANCES
2	CR	1.0000	CRANE
3	EL	1.0000	ELECTRICAL
4	EQ	1.0000	EQUIPMENT
5	HD	1.0000	HELIDECK
6	LA	1.0000	LIFT APPURTENANCES
7	MM	1.0000	MUD MATS
8	MS	1.0000	MISCELLANEOUS STUCTURAL STEEL
9	ST	1.0000	STRUCTURAL
10	MI	1.0000	MISCELLANEOUS

*** WEIGHT CONTROL ITEM DESCRIPTIONS ***

NO.	ITEM CODE	DISC. CODE	***** DESCRIPTION *****
1	RISE	AP	UNMODELED RISERS AND CLAMPS
2	COND	AP	UNMODELED CONDUCTORS
3	BOAT	AP	UNMODELED BOAT LANDINGS
4	BUMP	AP	UNMODELED BARGE BUMPERS
5	MISC	AP	MISCELLANEOUS APPURTENANCES
6	BOOM	CR	CRANE BOOM
7	BREST	CR	CRANE BOOM REST
8	CRANE	CR	CRANE, AND CRANE PEDESTAL
9	BLDG1	EQ	CONTROL & ENGINE BUILDING
10	SKID1	EQ	COMPRESSOR SKID
11	SKID2	EQ	FIRE EQUIPMENT SKID
12	MISC	EQ	MISCELLANEOUS EQUIPMENT
13	ELECT	EL	MISCELLANEOUS ELECTRICAL
14	STAIR	HD	HELIDECK STAIRS
15	SNET	HD	HELIDECK SAFETY NET
16	WALK	HD	HELIDECK WALKWAY
17	MISC	HD	HELIDECK MISCELLANEOUS
18	LEYE	LA	LIFTING EYES
19	SHACK	LA	LIFTING SHACKLES
20	SLING	LA	LIFTING SLINGS
21	MISC	LA	MISCELLANEOUS LIFT APPURTENANCES
22	MMAT	MM	MUD MAT PLATE
23	MSTIF	MM	MUD MAT STIFFENERS
24	DBEAM	MS	UNMODELED DECK BEAMS
25	GRATE	MS	GRATING
26	HRAIL	MS	HAND RAILING
27	PLATE	MS	UNMODELED PLATE
28	STAIR	MS	STAIRS
29	WALK	MS	WALKWAYS
30	MISC	MS	MISCELLANOUS UNMODELED ITEMS

*** WEIGHT AND CG DETAIL REPORT ***
CONFIGURATION:DECK LIFT CONFIGURATION

DISCIPLINE: CRANE											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
BREST	CRANE BOOM REST	1	0.00	0.00	2.000	1.050	2.000	2.100	-29.53	-9.84	19.68
CRANE	CRANE, AND CRANE PEDESTAL	1	0.00	0.00	10.000	1.050	10.000	10.500	-29.53	-29.53	19.68
TOTAL							12.000	12.600	-29.53	-26.25	19.68
DISCIPLINE: ELECTRICAL											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
ELECT	MISCELLANEOUS ELECTRICAL	6	0.00	0.00	9.685	1.050	58.110	61.016	0.00	0.00	21.19
TOTAL							58.110	61.016	0.00	0.00	21.19
DISCIPLINE: EQUIPMENT											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
BLDG1	CONTROL & ENGINE BUILDING	6	0.00	0.00	20.834	1.050	125.001	131.251	20.00	25.00	21.34
SKID1	COMPRESSOR SKID	5	0.00	0.00	15.000	1.050	75.000	78.750	20.00	5.00	21.26
SKID2	FIRE EQUIPMENT SKID	6	0.00	0.00	10.833	1.050	65.001	68.251	-20.00	-15.00	21.34
TOTAL							265.002	278.252	10.19	9.53	21.32
DISCIPLINE: LIFT APPURTENANCES											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
LEYE	LIFTING EYES	4	0.00	0.00	0.500	1.050	2.000	2.100	0.00	0.00	19.68
TOTAL							2.000	2.100	0.00	0.00	19.68
DISCIPLINE: MISCELLANEOUS STRUCTURAL STEEL											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
DBEAM	UNMODELED DECK BEAMS	12	0.00	0.00	21.791	1.050	261.496	274.571	0.00	0.00	21.18
HRAIL	HAND RAILING	6	0.00	0.00	0.295	1.050	1.772	1.860	-14.76	14.76	21.18
STAIR	STAIRS	2	0.00	0.00	3.000	1.050	6.000	6.300	29.53	-5.84	21.18
WALK	WALKWAYS	3	0.00	0.00	1.476	1.050	4.429	4.651	-29.53	0.00	21.18
TOTAL							273.697	287.382	0.07	-0.03	21.18
DISCIPLINE: STRUCTURAL											
ITEM	NO.	UNIT	WEIGHT	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***					
CODE	***** DESCRIPTION *****	REQ.	LENGTH	TOTAL	WEIGHT	NET	GROSS	X	Y	Z	
			FT	LENGTH	WEIGHT	NET	GROSS	FT	FT	FT	
				FT	K/FT	FACTOR	KIPS	KIPS			
DK1	W36X210	24	19.69	472.44	0.210	1.050	99.350	104.318	0.00	0.00	21.18
DK2	W24X131	8	20.88	167.03	0.131	1.050	21.883	22.977	0.00	0.00	21.68
PL3	TUB 29.9 X 0.55	4	22.97	91.86	0.173	1.050	15.892	16.687	0.00	0.00	8.20

*** WEIGHT AND CG DETAIL REPORT ***
 CONFIGURATION:DECK IN-PLACE CONFIGURATION

DISCIPLINE: CRANE											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
BREST	CRANE BOOM REST	1	0.00	0.00	2.000	1.050	2.000	2.100	-29.53	-9.84	19.68
CRANE	CRANE, AND CRANE PEDESTAL	1	0.00	0.00	10.000	1.050	10.000	10.500	-29.53	-29.53	19.68
TOTAL							12.000	12.600	-29.53	-26.25	19.68
DISCIPLINE: ELECTRICAL											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
ELECT	MISCELLANEOUS ELECTRICAL	6	0.00	0.00	9.685	1.050	58.110	61.016	0.00	0.00	21.19
TOTAL							58.110	61.016	0.00	0.00	21.19
DISCIPLINE: EQUIPMENT											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
BLDG1	CONTROL & ENGINE BUILDING	12	0.00	0.00	11.458	1.050	137.501	144.376	20.00	25.00	21.34
SKID1	COMPRESSOR SKID	10	0.00	0.00	8.250	1.050	82.500	86.625	20.00	5.00	21.26
SKID2	FIRE EQUIPMENT SKID	12	0.00	0.00	5.963	1.050	71.561	75.139	-20.01	-15.00	21.34
TOTAL							291.562	306.140	10.18	9.52	21.32
DISCIPLINE: LIFT APPURTENANCES											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
LEYE	LIFTING EYES	4	0.00	0.00	0.500	1.050	2.000	2.100	0.00	0.00	19.68
TOTAL							2.000	2.100	0.00	0.00	19.68
DISCIPLINE: MISCELLANEOUS STRUCTURAL STEEL											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
DBEAM	UNMODELED DECK BEAMS	12	0.00	0.00	21.791	1.050	261.496	274.571	0.00	0.00	21.18
HRAIL	HAND RAILING	6	0.00	0.00	0.295	1.050	1.772	1.860	-14.76	14.76	21.18
STAIR	STAIRS	2	0.00	0.00	3.000	1.050	6.000	6.300	29.53	-5.84	21.18
WALK	WALKWAYS	3	0.00	0.00	1.476	1.050	4.429	4.651	-29.53	0.00	21.18
TOTAL							273.697	287.382	0.07	-0.03	21.18
DISCIPLINE: STRUCTURAL											
ITEM	NO.	DISCIPLINE:	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***				
CODE	***** DESCRIPTION *****	REQ.	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z	
			FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT	
DK1	W36X210	24	19.69	472.44	0.210	1.050	99.350	104.318	0.00	0.00	21.18
DK2	W24X131	8	20.88	167.03	0.131	1.050	21.883	22.977	0.00	0.00	21.68
PL3	TUB 29.9 X 0.55	4	22.97	91.86	0.173	1.050	15.892	16.687	0.00	0.00	8.20

*** WEIGHT AND CG DETAIL REPORT ***
 CONFIGURATION: JACKET LIFT CONFIGURATION

DISCIPLINE: LIFT APPURTENANCES											
ITEM	***** DESCRIPTION *****	NO.	LENGTH	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***			
CODE		REQ.	FT	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z
			FT	FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT
LEVE	LIFTING EYES	4	0.00	0.00	0.500	1.050	2.000	2.100	0.00	0.00	-3.28
TOTAL							2.000	2.100	0.00	0.00	-3.28
DISCIPLINE: MUD MATS											
ITEM	***** DESCRIPTION *****	NO.	LENGTH	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***			
CODE		REQ.	FT	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z
			FT	FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT
MMAT	MUD MAT PLATE	4	0.00	0.00	3.000	1.050	12.000	12.600	0.00	0.00	-82.02
TOTAL							12.000	12.600	0.00	0.00	-82.02
DISCIPLINE: MISCELLANEOUS STRUCTURAL STEEL											
ITEM	***** DESCRIPTION *****	NO.	LENGTH	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***			
CODE		REQ.	FT	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z
			FT	FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT
HRAIL	HAND RAILING	2	0.00	0.00	0.295	1.050	0.591	0.620	4.92	-4.92	-3.28
STAIR	STAIRS	2	0.00	0.00	3.000	1.050	6.000	6.300	9.84	-5.84	-3.28
WALK	WALKWAYS	2	0.00	0.00	1.476	1.050	2.953	3.100	4.92	-4.92	-3.28
TOTAL							9.543	10.020	8.02	-5.50	-3.28
DISCIPLINE: STRUCTURAL											
ITEM	***** DESCRIPTION *****	NO.	LENGTH	TOTAL	UNIT	WEIGHT	** TOTAL WEIGHT **	*** CENTER OF GRAVITY ***			
CODE		REQ.	FT	LENGTH	WEIGHT	CONT.	NET	GROSS	X	Y	Z
			FT	FT	K/FT	FACTOR	KIPS	KIPS	FT	FT	FT
CON	TUB 19.7 X 0.55	7	15.68	109.77	0.225	1.050	24.749	25.987	0.00	0.00	-34.54
DB1	TUB 19.7 X 0.63	16	27.41	438.63	0.128	1.050	56.290	59.104	0.00	0.00	-60.70
DB2	TUB 15.0 X 0.55	16	21.84	349.47	0.085	1.050	29.662	31.145	0.00	0.00	-21.33
HB1	TUB 19.7 X 0.63	8	19.69	157.48	0.128	1.050	20.210	21.220	0.00	0.00	-82.02
HB2	TUB 15.0 X 0.55	4	28.71	114.83	0.085	1.050	9.747	10.234	0.00	0.00	-39.37
HB3	TUB 11.8 X 0.39	4	19.68	78.74	0.048	1.050	3.786	3.976	0.00	0.00	-3.28
HD1	TUB 15.0 X 0.55	8	23.76	190.10	0.085	1.050	16.135	16.942	0.00	0.00	-82.02
HD2	TUB 15.0 X 0.39	4	20.30	81.20	0.061	1.050	4.982	5.231	0.00	0.00	-39.37
HD3	TUB 11.8 X 0.39	4	13.92	55.68	0.048	1.050	2.677	2.811	0.00	0.00	-3.28
LG1	TUB 29.9 X 0.79	4	32.80	131.20	0.245	1.050	32.158	33.766	0.00	0.00	-65.87
LG1	TUB 29.9 X 0.71	4	10.51	42.05	0.221	1.050	9.310	9.775	0.00	0.00	-44.55
LG2	TUB 29.9 X 0.63	4	36.65	146.60	0.197	1.050	28.919	30.365	0.00	0.00	-21.33
PL4	TUB 29.9 X 0.79	8	3.58	28.65	0.245	1.050	7.023	7.375	0.00	0.00	-85.06
STB	TUB 23.6 X 0.63	0	0.00	0.00	0.000	1.050	0.000	0.000	0.00	0.00	0.00
VB1	TUB 19.7 X 0.63	4	24.86	99.43	0.128	1.050	12.760	13.398	0.00	0.00	-69.69
WSB	TUB 23.6 X 0.55	0	0.00	0.00	0.000	1.050	0.000	0.000	0.00	0.00	0.00
	NON-GROUP PLATES	0			0.000	1.050	0.000	0.000	0.00	0.00	0.00
TOTAL							258.408	271.329	0.00	0.00	-50.78

*** WEIGHT AND CG DETAIL REPORT ***
 CONFIGURATION:PILE CONFIGURATION

ITEM CODE	***** DESCRIPTION	***** REQ.	DISCIPLINE:		STRUCTURAL		UNIT WEIGHT K/FT	WEIGHT CONT. FACTOR	** TOTAL WEIGHT **		*** CENTER OF GRAVITY ***		
			NO. LENGTH FT	TOTAL LENGTH FT	NET KIPS	GROSS KIPS			X FT	Y FT	Z FT		
PL1	TUB 23.6 X 0.55	4	43.31	173.25	0.136	1.050	23.544	24.721	0.00	0.00	-60.70		
PL2	TUB 23.6 X 0.55	8	18.57	148.60	0.136	1.050	20.193	21.203	0.00	0.00	-21.57		
	NON-GRUP PLATES	0			0.000	1.050	0.000	0.000	0.00	0.00	0.00		
TOTAL							43.737	45.924	0.00	0.00	-42.63		

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*** WEIGHT AND CG SUMMARY REPORT ***

CONFIGURATION*				DECK LIFT CONFIGURATION			DECK IN-PLACE CONFIGURAT			JACKET LIFT CONFIGURATIO			**PILE		
DISC CODE	DISCIPLINE ***** DESCRIPTION	***** WEIGHT KIPS	***** FACTOR	***** WEIGHT KIPS	***** FACTOR	***** WEIGHT KIPS	***** WEIGHT KIPS	***** FACTOR	***** WEIGHT KIPS	***** FACTOR	***** WEIGHT KIPS	***** FACTOR	***** WEIGHT KIPS		
AP	APPURTENANCES	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00		
CR	CRANE	12.00	1.050	12.60	12.00	1.050	12.60	0.00	1.050	0.00	0.00	1.050	0.00		
EL	ELECTRICAL	58.11	1.050	61.02	58.11	1.050	61.02	0.00	1.050	0.00	0.00	1.050	0.00		
EQ	EQUIPMENT	265.00	1.050	278.25	291.56	1.050	306.14	0.00	1.050	0.00	0.00	1.050	0.00		
HD	HELIDECK	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00		
LA	LIFT APPURTENANCES	2.00	1.050	2.10	2.00	1.050	2.10	2.00	1.050	2.10	0.00	1.050	0.00		
MM	MUD MATS	0.00	1.050	0.00	0.00	1.050	0.00	12.00	1.050	12.60	0.00	1.050	0.00		
MS	MISCELLANEOUS STUCT	273.70	1.050	287.38	273.70	1.050	287.38	9.54	1.050	10.02	0.00	1.050	0.00		
ST	STRUCTURAL	417.45	1.050	438.33	417.45	1.050	438.33	258.41	1.050	271.33	43.74	1.050	45.92		
MI	MISCELLANEOUS	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00	0.00	1.050	0.00		
** TOTAL **		1028.26	1.050	1079.67	1054.82	1.050	1107.56	281.95	1.050	296.05	43.74	1.050	45.92		
CG LOCATION - X		2.30		2.30	2.50		2.50	0.27		0.27	0.00		0.00		
CG LOCATION - Y		2.14		2.14	2.33		2.33	-0.19		-0.19	0.00		0.00		
CG LOCATION - Z		20.60		20.60	20.62		20.62	-50.17		-50.17	-42.63		-42.63		

7 INPUT LINES

ANODE GROUP LABEL LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR ANODE CALCULATION PER NACE SP0176-2007 (FORMERLY RPO176-2003) AND DNV-RP-B401 SPECIFICATIONS.
(8-10)	ENTER THE ANODE GROUP LABEL
(12-17)	ENTER THE ORIGINAL ANODE LENTH.
(18-23)	ENTER THE ORIGINAL ANODE CROSS-SECTIONAL AREA (INCLUDING CORE).
(24-29)	ENTER THE ANODE CORE RADIUS.
(30-35)	ENTER THE ANODE WEIGHT.
(36-41)	ENTER THE ENERGY CAPABILITY (CURRENT CAPACITY) OF THE ANODE.
(42-47)	ENTER THE NET DRIVING POTENTIAL FOR THE ANODE. THE NET DRIVING POTENTIAL IS THE DIFFERENCE BETWEEN THE STRUCTURE'S MINIMUM PROTECTIVE POTENTIAL AND THE ANODES POTENTIAL.
(48-53)	ENTER THE ANODE'S UTILIZATION FACTOR OR EFFICIENCY USED TO CALCULATE THE EXPENDED ANODE DIMENSIONS. FOR STANDOFF-TYPE ANODES, UTILIZATION FACTORS OF 0.90-0.95 ARE COMMON. FOR BRACELET ANODES, VALUES OF 0.75-0.90 MAY BE USED.
(54-59)	ENTER THE PERCENTAGE OF THE ANODE'S LENGTH REMAINING AT THE END OF LIFE.
(60-65)	ENTER THE COST PER ANODE.

LINE LABEL	ANODE GROUP LABEL	SIZE				ENERGY CAPABILITY	NET DRIVING POTENTIAL	UTILIZATION FACTOR	% OF LENGTH AT END OF LIFE	ANODE COST	LEAVE BLANK
		LENGTH	AREA	CORE RADIUS	WEIGHT						
AGRUP											
1-- 5	8--10	12--17	18--23	24--29	30--35	36--41	42--47	48--53	54--59	60--65	66--80
DEFAULT									1		
ENGLISH		INCHES	SQ. IN	INCHES	LBS	A-H/LB	VOLTS			\$/ANODE	
METRIC		CM	SQ. CM	CM	KG	A-H/KG	VOLTS			\$/ANODE	

ANODE LEVEL LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED FOR THE ANODE CALCULATION PER NACE SP0176-2007 (FORMERLY RPO176-2003) AND DNV-RP-B401 SPECIFICATIONS. LEVELS MUST BE ENTERED IN INCREASING ORDER.
(8-10)	ENTER THE ANODE GROUP LABEL FOR THIS LEVEL.
(12-18)	ENTER THE DISTANCE FROM THE WATER LINE TO THE TOP DEPTH OF THIS LEVEL. A NEGATIVE VALUE INDICATES THE LEVEL IS ABOVE THE WATER LINE. IF THIS IS THE FIRST LEVEL AND NO VALUE IS GIVEN, THEN THE WATER LINE WILL BE USED. IF THIS IS NOT THE FIRST LEVEL AND NO VALUE IS GIVEN, THEN THE BOTTOM LEVEL OF THE LEVEL ABOVE WILL BE USED.
(19-25)	ENTER THE DISTANCE FROM THE WATER LINE TO THE BOTTOM DEPTH OF THIS LEVEL
(27-33)	ENTER THE INITIAL CURRENT DENSITY REQUIRED TO POLARIZE THIS LEVEL. USED IN NACE APPENDIX D / DNV ANODE CALCULATIONS.
(34-40)	ENTER THE MEAN CURRENT DENSITY REQUIRED TO MAINTAIN THE PROTECTIVE CURRENT TO THIS LEVEL OVER THE DESIGN LIFE. USED IN NACE APPENDIX D / DNV AND NACE APPENIX E ANODE CALCULATIONS.
(41-47)	ENTER THE FINAL CURRENT DENSITY REQUIRED TO PROVIDE THE PROTECTIVE CURRENT TO THIS LEVEL AT THE END OF THE DESIGN LIFE . USED IN NACE APPENDIX D / DNV ANODE CALCULATIONS.
(48-54)	ENTER THE WATER RESISTIVITY OF THIS LEVEL. USED IN NACE APPENDIX D / DNV AND NACE APPENDIX E ANODE CALCULATIONS.
(55-61)	ENTER THE SURFACE AREA FACTOR OR SAFETY FACTOR TO BE APPLIED TO THIS LEVEL. USED IN NACE APPENDIX D / DNV AND NACE APPENDIX E ANODE CALCULATIONS.

LINE LABEL	ANODE GROUP LABEL	TOP DEPTH	BOTTOM DEPTH	INITIAL CURRENT DENSITY	MEAN CURRENT DENSITY	FINAL CURRENT DENSITY	WATER RESISTIVITY	SURFACE AREA FACTOR	LEAVE BLANK
ANLEV									
1-- 5	8--10	12--18	19--25	27--33	34--40	41--47	48--54	55--61	62--80
DEFAULT				10.0 ENGL.	5.0 ENGL.	7.0 ENGL.	0.66 ENGL.	1	
ENGLISH		FT	FT	MA/SQ.FT.	MA/SQ.FT.	MA/SQ.FT.	OHM-FT		
METRIC		M	M	MA/M**2	MA/M**2	MA/M**2	OHM-M		

ANODE DATA LINE

COLUMNS	COMMENTARY
LOCATION	THIS LINE FOLLOWS THE MTOPT LINE.
GENERAL	THIS LINE ENABLES THE USE TO SPECIFY THE ANODE REQUIREMENTS VERSUS WATER DEPTH. THIS DATA MUST BE IN ORDER OF INCREASING WATER DEPTH AND REQUIRES A HEADER LINE. THE ANODE SIZE AND MATERIAL COST SHOULD BE ENTERED ON THE FIRST NON-HEADER ANODE LINE.
(12-20)	ENTER THE WATER DEPTH TO WHICH THIS ANODE REQUIREMENT IS VALID. THE SURFACE AREA AND NUMBER OF ANODES WILL BE COMPUTED FOR EACH DEPTH INCREMENT.
(21-30)	ENTER THE AMOUNT OF ANODE MATERIAL REQUIRED TO PROVIDE A UNIT SURFACE AREA WITH PROTECTION THROUGHOUT THE DESIGN LIFE OF THE STRUCTURE.
(31-40)	ENTER THE ANODE SIZE ONLY ON THE FIRST (NON-HEADER) ANODE LINE.
(41-50)	ENTER THE ANODE MATERIAL COST ONLY ON THE FIRST (NON-HEADER) ANODE LINE.
(51-55)	ENTER THE PLATE AREA FACTOR ONLY ON THE FIRST (NON-HEADER) ANODE LINE. THIS FACTOR SHOULD BE 1.0 FOR ALL PLATES EXPOSED TO SEAWATER ONLY ON ONE SIDE AND SHOULD BE 2.0 FOR ALL PLATES EXPOSED ON BOTH SIDES.

LINE LABEL	DEPTH	ANODE REQUIREMENT	ANODE SIZE	ANODE MATERIAL COST	PLATE AREA FACTOR	LEAVE BLANK
ANODE						
1-- 5	12<--20	21<--30	31<--40	41<--50	51<--55	56----80
DEFAULT					1	
ENGLISH	FT	LB/SQ.FT	LB	\$/LB		
METRIC	M	KG/SQ.M	KG	\$/KG		

ADDITIONAL ANODE OPTIONS LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	THIS LINE FOLLOWS THE MTOPT LINE.
GENERAL	THIS LINE ENABLES THE CALCULATION OF ANODE REQUIREMENT FOLLOWING NACE SP0176-2007 (FORMERLY RPO176-2003) AND DNV-RP-B401 SPECIFICATIONS.
(7- 8)	ENTER THE ANODE CALCULATIN OPTION: ND - USE NACE APPENDIX D / DNV NE - USE NACE APPENDIX E NB - USE BOTH NACE APPENDIX D / DNV AND NACE APPENDIX E SA - USE SACS ANODE METHOD (DEFAULT)
(10-15)	ENTER THE DESIGN LIFE TO BE USED IN NACE APPENDIX D / DNV ANODE CALCULATIONS.
(17-22)	ENTER THE TYPICAL DESIGN SLOPE TO BE USED IN NACE APPENDIX E ANODE CALCULATIONS.
(24)	ENTER THE DESIRED REPORT OPTIONS: 0 - SUMMARY PRINT (DEFAULT) 1 - FULL PRINT
(26-27)	ENTER THE NUMBER OF WELLS/CONDUCTORS TO BE USED TO CALCULATE ADDITIONAL CURRENT DRAIN IMPOSED BY THE WELL CASINGS BELOW THE MUDLINE.
(29-35)	ENTER THE CURRENT DRAIN PER WELL/CONDUCTOR TO BE USED TO CALCULATE ADDITIONAL CURRENT DRAIN IMPOSED BY THE WELL CASINGS BELOW THE MUDLINE.
(37-42)	ENTER ANY ADDITIONAL CURRENT DRAIN TO BE USED.
(44-45)	ENTER 'RD' TO ROUND UP THE NUMBER OF ANODES CALCULATED IN THE ANODE SUMMARY REPORT TO THE NEAREST WHOLE NUMBER.
NOTE:	ADDITIONAL CURRENT DRAIN IS USED IN THE NACE APPENDIX D / DNV AND NACE APPENDIX E ANODE CALCULATIONS. ANODE PROPERTIES FROM THE LAST ANODE GROUP ENTERED WILL BE USED.

LINE LABEL	ANODE CALCULATION OPTION	DESIGN LIFE	TYPICAL DESIGN SLOPE	REPORT OPTIONS	NUMBER OF WELLS	CURRENT DRAIN PER WELL	ADDITIONAL CURRENT DRAIN	ANODE SUMMARY OPTION	LEAVE BLANK
ANOPT									
1-- 5	7<-- 8	10<--15	17<--22	24	26<--27	29<--35	37<--42	44<--45	46--80
DEFAULT	SA		44.0 ENGL.	0=		3			
ENGLISH		YEARS	OHM-SQ.FT.			AMPS/WELL	AMPS		
METRIC		YEARS	OHM-M**2			AMPS/WELL	AMPS		

CONFIGURATION DESCRIPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO IDENTIFY AND DESCRIBE EACH CONFIGURATION. WEIGHT CONTROL REPORTS ARE GENERATED FOR EACH CONFIGURATION DEFINED BY A 'CONFIG' LINE.
(7-12)	ENTER THE WEIGHT CONTINGENCY FACTOR. THIS FACTOR WILL BE USED TO MULTIPLY EVERY ELEMENT AND EVERY LOAD CASE ASSOCIATED WITH THIS CONFIGURATION.
(14)	ENTER 'I' IF ONLY LOAD CASES LISTED ON THE SUBSEQUENT 'LCASE' LINE ARE TO BE INCLUDED IN THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(16)	ENTER 'I' IF ONLY MEMBER GROUPS LISTED ON THE SUBSEQUENT 'GRPLST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(17)	ENTER 'I' IF ONLY MEMBERS LISTED ON THE SUBSEQUENT 'MBRLST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(18)	ENTER 'I' IF ONLY PLATE GROUPS LISTED ON THE SUBSEQUENT 'PGRPLST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(19)	ENTER 'I' IF ONLY PLATES LISTED ON THE SUBSEQUENT 'PLALST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(20)	IF THE PLATES WITHOUT GROUP IDENTIFIERS ARE TO BE INCLUDED IN THIS CONFIGURATION, ENTER 'I', OR TO EXCLUDE THEM ENTER 'X'.
(21)	ENTER 'I' IF ONLY PILE GROUPS LISTED ON THE SUBSEQUENT 'PILLST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION., OR ENTER 'X' TO EXCLUDE THEM.
(22)	ENTER 'I' IF ONLY DISCIPLINES LISTED ON THE SUBSEQUENT 'DISLST' LINE ARE TO BE CONSIDERED PART OF THIS CONFIGURATION, OR ENTER 'X' TO EXCLUDE THEM.
(23)	ENTER 'B' IF BOTH TUBULARS AND GROUT OF THE CONCENTRIC TUBULARS ARE BE INCLUDED IN WEIGHT CALCULATION. ENTER 'O' FOR OUTSIDE TUBULAR ONLY. ENTER 'I' FOR INSIDE TUBULAR ONLY. ENTER 'N' FOR BOTH TUBULARS EXCLUDING GROUT ARE TO BE INCLUDED.
(40-80)	ENTER THE DESCRIPTION OR NAME OF THIS CONFIGURATION.

LINE LABEL	WEIGHT CONTINGENCY FACTOR	LOAD CASE INCLUSION OPTION	MEMBER GRUP INCLUSION OPTION	MEMBER INCLUSION OPTION	PLATE GROUP INCLUSION OPTION	PLATE INCLUSION OPTION	NON-GROUPED PLATE INCLUSION OPTION	PILE GROUP INCLUSION OPTION	DISCIPLINE INCLUSION OPTION	CONCENTRIC TUBULAR OPTION	CONFIGURATION DESCRIPTION
CONFIG											
1-- 6	7<--12	14	16	17	18	19	20	21	22	23	40--80
DEFAULT	1	'I'	'I'	'I'	'I'	'I'	'I'	'I'	'X'	'B'	

COST OVERRIDE BY YIELD LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO OVERRIDE THE MATERIAL COST OF MEMBERS, PLATES, SHELLS, AND PILES BY THE SPECIFIED YIELD STRESS.
NOTE:	A HEADER LINE IS OPTIONAL
(10-15)	ENTER YIELD STRESS OF THE ELEMENT.
(16-22)	ENTER MATERIAL COST OVERRIDE PER UNIT WEIGHT.
NOTE:	2ND THROUGH 4TH OVERRIDES ARE SIMILAR

LINE LABEL	1ST OVERRIDE		2ND OVERRIDE		3RD OVERRIDE		4TH OVERRIDE		LEAVE BLANK
	YIELD	COST	YIELD	COST	YIELD	COST	YIELD	COST	
CSTYLD									
1-- 6	10--15	16--22	24--29	30--36	38--43	44--50	52--57	58--64	65--80
DEFAULT									
ENGLISH	KSI	\$/LB	KSI	\$/LB	KSI	\$/LB	KSI	\$/LB	
METRIC (KN)	KN/SQ.CM	\$/KG	KN/SQ.CM	\$/KG	KN/SQ.CM	\$/KG	KN/SQ.CM	\$/KG	
METRIC (KG)	KG/SQ.CM	\$/KG	KG/SQ.CM	\$/KG	KG/SQ.CM	\$/KG	KG/SQ.CM	\$/KG	

DISCIPLINE DESCRIPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE MAY BE INCLUDED IN THE MATERIAL TAKE-OFF INPUT FILE OR MAY BE CONTAINED IN AN AUXILIARY LIBRARY FILE. EACH WEIGHT CONTROL CONFIGURATION IS MADE UP OF DISCIPLINES. A DISCIPLINE CONSIST OF STRUCTURAL ELEMENTS AND/OR ONE OR MORE ITEMS WHOSE WEIGHT AND LOCATION ARE DEFINED BY 'LOAD' LINES IN THE SACS IV MODEL FILE.
(6- 7)	ENTER THE DISCIPLINE CODE FOR THIS DISCIPLINE. THIS TWO CHARACTER CODE IS REFERENCED ON THE ITEM LINE OF ITEMS THAT ARE ASSIGNED TO THIS DISCIPLINE. IT IS ALSO REFERENCED BY 'LOAD' LINES IN THE SACS IV MODEL FILE IN COLUMNS 73 AND 74.
(8-14)	ENTER THE WEIGHT CONTINGENCY FACTOR. THIS FACTOR WILL BE USED TO MULTIPLY EVERY ITEM ASSOCIATED WITH THIS DISCIPLINE.
(40-80)	ENTER THE DESCRIPTION OF THIS DISCIPLINE.

LINE LABEL	DISCIPLINE CODE	WEIGHT CONTINGENCY FACTOR	DISCIPLINE DESCRIPTION
DISC			
1-- 4	6-- 7	8<--14	40-----80
DEFAULT		1	

DISCIPLINE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE DISCIPLINES EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(8-42)	ENTER THE DISCIPLINES TO BE SELECTED. USE AS MANY DISLST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED DISCIPLINES FOR THIS CONFIGURATION.

LINE LABEL	SELECTED DISCIPLINES											
	1ST DISCIPLINE	2ND DISCIPLINE	3RD DISCIPLINE	4TH DISCIPLINE	5TH DISCIPLINE	6TH DISCIPLINE	7TH DISCIPLINE	8TH DISCIPLINE	9TH DISCIPLINE	10TH DISCIPLINE	11TH DISCIPLINE	12TH DISCIPLINE
DISLST												
1-- 6	8-- 9	11--12	14--15	17--18	20--21	23--24	26--27	29--30	32--33	35--36	38--39	41--42

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
----------------	-------------------

GENERAL THIS LINE IS OPTIONAL AND SIGNALS THE END OF THE CALCULATIONS.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

MEMBER GROUP SELECTION LINE

COLUMNS COMMENTARY

GENERAL THIS LINE IS USED TO SELECT THE MEMBER GROUPS EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.

(8-78) ENTER THE MEMBER GROUPS TO BE SELECTED. USE AS MANY GRPLST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED MEMBER GROUPS FOR THIS CONFIGURATION.

LINE LABEL	SELECTED MEMBER GROUPS																	
	1ST MEMBER GROUP	2ND MEMBER GROUP	3RD MEMBER GROUP	4TH MEMBER GROUP	5TH MEMBER GROUP	6TH MEMBER GROUP	7TH MEMBER GROUP	8TH MEMBER GROUP	9TH MEMBER GROUP	10TH MEMBER GROUP	11TH MEMBER GROUP	12TH MEMBER GROUP	13TH MEMBER GROUP	14TH MEMBER GROUP	15TH MEMBER GROUP	16TH MEMBER GROUP	17TH MEMBER GROUP	18TH MEMBER GROUP
GRPLST																		
1-- 6	8--10	12--14	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74	76--78

ITEM DESCRIPTION LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE MAY BE INCLUDED IN THE MATERIAL TAKE-OFF INPUT FILE OR MAY BE IN AN AUXILIARY LIBRARY FILE. 'LOAD' LINES IN THE SACS IV MODEL FILE DESCRIBE THE WEIGHT AND LOCATION OF ITEMS. LIKE ITEMS MAY BE GROUPED INTO DISCIPLINES.
(6-10)	ENTER THE ITEM CODE ASSIGNED TO THIS ITEM. ALL WEIGHTS DEFINED BY 'LOAD' LINES WITH THIS CODE SPECIFIED IN COLUMNS 76-80 WILL BE CONSIDERED AS PART OF THIS ITEM.
(14-15)	ENTER THE DISCIPLINE CODE OF THE DISCIPLINE TO WHICH THIS ITEM IS ASSIGNED. LEAVE BLANK IF THIS ITEM CAN BE USED BY MORE THAN ONE DISCIPLINE.
(40-80)	ENTER THE ITEM NAME OR DESCRIPTION.

LINE LABEL	ITEM CODE	DISCIPLINE CODE	ITEM DESCRIPTION
ITEM			
1-- 4	6--10	14--15	40-----80

LOAD CASE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE LOAD CASES EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(7-80)	ENTER THE LOAD CASES TO BE SELECTED. USE AS MANY LCASE RECORDS AS NECESSARY TO SELECT ALL THE DESIRED LOAD CASES FOR THIS CONFIGURATION.

LINE LABEL	SELECTED LOAD CASES														
	1ST LOAD CASE	2ND LOAD CASE	3RD LOAD CASE	4TH LOAD CASE	5TH LOAD CASE	6TH LOAD CASE	7TH LOAD CASE	8TH LOAD CASE	9TH LOAD CASE	10TH LOAD CASE	11TH LOAD CASE	12TH LOAD CASE	13TH LOAD CASE	14TH LOAD CASE	15TH LOAD CASE
LCASE															
1-- 5	7--10	12--15	17--20	22--25	27--30	32--35	37--40	42--45	47--50	52--55	57--60	62--65	67--70	72--75	77--80

MEMBER SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE MEMBERS EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(8-76)	ENTER THE MEMBERS TO BE SELECTED. USE AS MANY MBRLST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED MEMBERS FOR THIS CONFIGURATION.

LINE LABEL	SELECTED MEMBERS													
	1ST		2ND		3RD		4TH		5TH		6TH		7TH	
	JOINT A	JOINT B	JOINT A	JOINT B	JOINT A	JOINT B	JOINT A	JOINT B	JOINT A	JOINT B	JOINT A	JOINT B	JOINT A	JOINT B
MBRLST														
1-- 6	8--11	13--16	18--21	23--26	28--31	33--36	38--41	43--46	48--51	53--56	58--61	63--66	68--71	73--76

MEMBER OVERRIDE LINE

COMMENTARY

GENERAL THIS LINE IS USED TO EXCLUDE MEMBERS FROM ALL
CALCULATIONS, AND TO OVERRIDE MATERIAL COST AND SURFACE AREAS.

```
( 8 )      ENTER OVERRIDE OPTION:
           'X' TO EXCLUDE THESE MEMBERS FROM ALL CALCULATIONS.
           'I' TO INCLUDE ONLY THESE MEMBERS FOR ALL CALCULATIONS.
           'S' TO INCLUDE THESE MEMBERS FOR INTERNAL AND EXTERNAL
              SURFACE AREA CALCULATIONS.
           'A' TO EXCLUDE THESE MEMBERS FROM SURFACE AREA CALCULATIONS
              ONLY.
```

NOTE: INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND SHOULD NOT BE MIXED ON OTHER MBROVR LINES.

(10-68) ENTER ALL MEMBERS USED.

(70-75) ENTER COST OVERRIDE.

[illegible]

MEMBER GROUP OVERRIDE LINE

COLUMNS	COMMENTARY
---------	------------

GENERAL THIS LINE IS USED TO EXCLUDE MEMBER GROUPS FROM ALL
CALCULATIONS, AND TO OVERRIDE MATERIAL COST AND SURFACE AREAS.

```
( 8 )      ENTER OVERRIDE OPTION:
            'X' TO EXCLUDE THESE MEMBER GROUPS FROM ALL CALCULATIONS.
            'I' TO INCLUDE ONLY THESE MEMBER GROUPS FOR ALL CALCULATIONS.
            'S' TO INCLUDE THESE MEMBER GROUPS FOR INTERNAL AND
                EXTERNAL SURFACE AREA CALCULATIONS.
            'A' TO EXCLUDE THESE MEMBERS GROUPS FROM SURFACE AREA
                CALCULATIONS ONLY.
```

NOTE: INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND SHOULD NOT BE MIXED ON OTHER MGPOVR LINES.

(10-68) ENTER ALL MEMBER GROUPS USED.

(70-75) ENTER COST OVERRIDE.

[illegible]

MATERIAL TAKE-OFF OPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY MATERIAL TAKE-OFF ANALYSIS. IT IS USED TO SPECIFY THE OVERALL ANALYSIS PARAMETERS.
(9-10)	ENTER THE UPWARD COORDINATE DIRECTION USED IN THE SACS IV ANALYSIS. OPTIONS ARE + OR - X, Y OR Z.
(13)	ENTER THE PROCEDURE USED TO DETERMINE THE MEMBER LENGTH: '0' - USE INPUT JOINT THICKNESS ONLY. '1' - COMPUTE THICKNESS AND ADD TO INPUT JOINT THICKNESS. '2' - COMPUTE THICKNESS AND IGNORE INPUT JOINT THICKNESS. '3' - USE MEMBER OFFSETS ONLY AND IGNORE INPUT JOINT THICKNESS.
(15-16)	ENTER THE DESIRED PRINT LEVEL: 'SM' - SUMMARY PRINT. 'FL' - FULL PRINT. 'DG' - DIAGNOSTIC PRINT.
(18-19)	SELECT WELD VOLUME CALCULATION OPTION: 'WV' - EXECUTE WELD VOLUME CALCULATION. 'FL' - WELD VOLUME CALCULATION WITH FULL PRINT. LEAVE BLANK FOR NO WELD VOLUME CALCULATION.
(21-22)	ENTER 'CS' TO EXECUTE COST ANALYSIS.
(24-25)	ENTER 'SA' TO EXECUTE SURFACE AREA CALCULATION (REQUIRES FULL PRINT OPTION).
(30-36)	ENTER THE WATER DEPTH.
(37-43)	ENTER THE VERTICAL COORDINATE OF THE MUDLINE ELEVATION.
(44-50)	ENTER THE MATERIAL DENSITY OF THE STRUCTURE.
(51-57)	ENTER THE MATERIAL COST. EXCEPTIONS TO THIS COST CAN BE SPECIFIED ON THE 'GRPCST' AND 'PGPCST' INPUT LINES.
(65-71)	ENTER THE WELD VOLUME COST.
(72)	ENTER 'I' TO INCLUDE PILE ELEMENTS, OR 'E' TO EXCLUDE PILE ELEMENTS.
NOTE:	COST INFORMATION CAN BE INPUT IN ANY MONETARY SYSTEM. THE SYMBOL \$ IS USED TO DESIGNATE A UNIT OF THAT MONETARY SYSTEM.

LINE LABEL	VERTICAL COORDINATE	OUTPUT OPTIONS					GENERAL PARAMETERS			COST DATA		INCLUDE PILE ELEMENTS	LEAVE BLANK
		MEMBER END OPTION	PRINT LEVEL	WELD VOLUME OPTION	COST ANALYSIS OPTION	SURFACE AREA OPTION	WATER DEPTH	MUDLINE ELEV.	MATERIAL DENSITY	MATERIAL COST	WELD COST		
MTOPT													
1-- 5	9--10	13	15--16	18--19	21--22	24--25	30<--36	37<--43	44<--50	51<--57	65<--71	72	73--80
DEFAULT	+Z	0	FL									E	
ENGLISH							FT	FT	LB/CU.FT	\$/LB	\$/CU.IN		
METRIC							M	M	TONNE/CU.M	\$/KG	\$/CU.CM		

SECOND MATERIAL TAKE-OFF OPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY MATERIAL TAKE-OFF RUN. IT IS USED TO SPECIFY THE ADDITIONAL OVERALL ANALYSIS PARAMETERS.
(9-15)	ENTER THE DENSITY OF CONCRETE.
(16-22)	ENTER THE COST OF CONCRETE.
(24)	ENTER 'I' TO INCLUDE THE WEIGHT OF CONCRETE SHEAR REINFORCEMENT , OR 'X' TO EXCLUDE.
NOTE:	ALL COST INFORMATION CAN BE INPUT IN ANY MONETARY SYSTEM. THE SYMBOL \$ IS USED TO DESIGNATE A UNIT OF THAT MONETARY SYSTEM.

LINE LABEL	CONCRETE			LEAVE BLANK
	DENSITY	COST	INCLUDE SHEAR REINFORCEMENT	
MTOPT2				
1-- 6	9<--15	16<--22	24	25-----80
DEFAULT	145.0 ENGL		I	
ENGLISH	LB/CU.FT	\$/LB		
METRIC	TONNE/CU.M	\$/KG		

OBJECT DESCRIPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE MAY BE INCLUDED IN THE MATERIAL TAKE-OFF INPUT FILE OR MAY BE IN AN AUXILIARY LIBRARY FILE. 'LOAD' LINES IN THE SACS IV MODEL FILE DESCRIBE THE WEIGHT AND LOCATION OF OBJECT. LIKE OBJECT MAY BE GROUPED INTO DISCIPLINES.
(8)	ENTER 'W' IF OBJECT TYPE IS A USER DEFINED WEIGHT. 'L' IF OBJECT TYPE IS A USER DEFINED LOAD.
(10-13)	ENTER THE OBJECT ID NAME.
(15-18)	ENTER THE LOAD CASE OF THIS OBJECT, OR WEIGHT GROUP ID IF OBJECT TYPE IS A USER DEFINED WEIGHT.
(20-27)	ENTER THE LOAD LABEL OF THIS OBJECT, OR WEIGHT ID IF OBJECT TYPE IS A USER DEFINED WEIGHT.
(29-35)	ENTER THE LOAD FACTOR FOR THIS OBJECT.
(40-80)	ENTER THE OBJECT NAME OR DESCRIPTION.
NOTE:	SPECIAL CHARACTERS * AND ? CAN BE USED IN THE LOAD CASE AND LOAD LABEL ENTRY. THE * REFERS TO ANY SINGLE OR SET OF CHARACTERS. THE ? REFERS TO ANY SINGLE CHARACTER.

LINE LABEL	OBJECT TYPE	OBJECT ID	LOAD CASE OR WEIGHT GROUP ID	LOAD LABEL OR WEIGHT ID	OBJECT FACTOR	OBJECT DESCRIPTION
OBJECT						
1-- 6	8	10--13	15--18	20--27	29--35	40-----80
DEFAULT	L				1	

OBJECT INCLUSION LINE

COLUMNS COMMENTARY

GENERAL THIS LINE IS USED TO INCLUDE OBJECTS IN A DISCIPLINE.

(8-76) ENTER THE OBJECTS INCLUDED IN THE DISCIPLINE. USE AS MANY
OBJINC RECORDS AS NECESSARY TO SELECT ALL THE DESIRED OBJECTS
FOR THIS DISCIPLINE.

NOTE: THIS LINE MUST FOLLOW A DISCIPLINE DESCRIPTION LINE.

LINE LABEL	INCLUDED OBJECTS													
	1ST OBJECT	2ND OBJECT	3RD OBJECT	4TH OBJECT	5TH OBJECT	6TH OBJECT	7TH OBJECT	8TH OBJECT	9TH OBJECT	10TH OBJECT	11TH OBJECT	12TH OBJECT	13TH OBJECT	14TH OBJECT
OBJINC														
1-- 6	8--11	13--16	18--21	23--26	28--31	33--36	38--41	43--46	48--51	53--56	58--61	63--66	68--71	73--76

PLATE GROUP OVERRIDE LINE

COLUMNS	COMMENTARY
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GENERAL THIS LINE IS USED TO EXCLUDE PLATE GROUPS FROM ALL
CALCULATIONS, AND TO OVERRIDE MATERIAL COST AND SURFACE AREAS.

```
( 8 )      ENTER OVERRIDE OPTION:
            'X' TO EXCLUDE THESE PLATE GROUPS FROM ALL CALCULATIONS.
            'I' TO INCLUDE ONLY THESE PLATE GROUPS FOR ALL CALCULATIONS.
            'A' TO EXCLUDE THESE PLATE GROUPS FROM SURFACE AREA
                CALCULATIONS ONLY.
```

NOTE: INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND SHOULD NOT BE MIXED ON OTHER PGPOVR LINES.

(10-60) ENTER ALL PLATE GROUPS USED.

(63-68) ENTER PLATE SURFACE AREA FACTOR TO BE USED IN ANODE CALCULATIONS. THIS FACTOR SHOULD BE 1.0 FOR ALL PLATES EXPOSED TO SEAWATER ONLY ON ONE SIDE AND SHOULD BE 2.0 FOR ALL PLATES EXPOSED ON BOTH SIDES.

(70-75) ENTER COST OVERRIDE.

[illegible]

PLATE GROUP SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE PLATE GROUPS EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(8-78)	ENTER THE PLATE GROUPS TO BE SELECTED. USE AS MANY PGRPLST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED PLATE GROUPS FOR THIS CONFIGURATION.

LINE LABEL	SELECTED PLATE GROUPS																	
	1ST PLATE GROUP	2ND PLATE GROUP	3RD PLATE GROUP	4TH PLATE GROUP	5TH PLATE GROUP	6TH PLATE GROUP	7TH PLATE GROUP	8TH PLATE GROUP	9TH PLATE GROUP	10TH PLATE GROUP	11TH PLATE GROUP	12TH PLATE GROUP	13TH PLATE GROUP	14TH PLATE GROUP	15TH PLATE GROUP	16TH PLATE GROUP	17TH PLATE GROUP	18TH PLATE GROUP
PGRPLST																		
1-- 7	8--10	12--14	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74	76--78

PILE GROUP SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE PILE GROUPS DEFINED IN A PSI INPUT FILE TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(8-78)	ENTER THE PILE GROUPS TO BE SELECTED. USE AS MANY PILLST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED PILE GROUPS FOR THIS CONFIGURATION.

LINE LABEL	SELECTED PILE GROUPS																	
	1ST PILE GROUP	2ND PILE GROUP	3RD PILE GROUP	4TH PILE GROUP	5TH PILE GROUP	6TH PILE GROUP	7TH PILE GROUP	8TH PILE GROUP	9TH PILE GROUP	10TH PILE GROUP	11TH PILE GROUP	12TH PILE GROUP	13TH PILE GROUP	14TH PILE GROUP	15TH PILE GROUP	16TH PILE GROUP	17TH PILE GROUP	18TH PILE GROUP
PILLST																		
1-- 6	8--10	12--14	16--18	20--22	24--26	28--30	32--34	36--38	40--42	44--46	48--50	52--54	56--58	60--62	64--66	68--70	72--74	76--78

PILE GROUP OVERRIDE LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO EXCLUDE PILE GROUPS DEFINED IN A PSI INPUT FILE FROM ALL CALCULATIONS, AND TO OVERRIDE MATERIAL COST AND SURFACE AREAS.
(8)	ENTER OVERRIDE OPTION: 'X' TO EXCLUDE THESE PILE GROUPS FROM ALL CALCULATIONS. 'I' TO INCLUDE ONLY THESE PILE GROUPS FOR ALL CALCULATIONS. 'S' TO INCLUDE THESE PILE GROUPS FOR INTERNAL AND EXTERNAL SURFACE AREA CALCULATIONS. 'A' TO EXCLUDE THESE PILE GROUPS FROM SURFACE AREA CALCULATIONS ONLY. NOTE: INCLUDE AND EXCLUDE OPTIONS ARE MUTUALLY EXCLUSIVE AND SHOULD NOT BE MIXED ON OTHER PILOVR LINES.
(10-68)	ENTER ALL PILE GROUPS USED.
(76-80)	ENTER COST OVERRIDE.

[illegible]

PLATE SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE PLATES EITHER TO BE INCLUDED OR EXCLUDED IN THIS CONFIGURATION.
(8-76)	ENTER THE PLATES TO BE SELECTED. USE AS MANY PLALST RECORDS AS NECESSARY TO SELECT ALL THE DESIRED PLATES FOR THIS CONFIGURATION.

LINE LABEL	SELECTED PLATES													
	1ST PLATE	2ND PLATE	3RD PLATE	4TH PLATE	5TH PLATE	6TH PLATE	7TH PLATE	8TH PLATE	9TH PLATE	10TH PLATE	11TH PLATE	12TH PLATE	13TH PLATE	14TH PLATE
PLALST														
1-- 6	8--11	13--16	18--21	23--26	28--31	33--36	38--41	43--46	48--51	53--56	58--61	63--66	68--71	73--76

PLATE OVERRIDE LINE

COLUMNS

COMMENTARY

GENERAL THIS LINE IS USED TO EXCLUDE PLATES FROM ALL CALCULATIONS,
AND TO OVERRIDE MATERIAL COST AND SURFACE AREAS.

(8)

ENTER OVERRIDE OPTION:

'X' TO EXCLUDE THESE PLATES FROM ALL CALCULATIONS.

'I' TO INCLUDE ONLY THESE PLATES FOR ALL CALCULATIONS.

'A' TO EXCLUDE THESE PLATES FROM SURFACE AREA CALCULATIONS

NOTE: INCLUDE AND EXCLUDE OPTIONS ARE MUTUALY EXCLUSIVE AND SHOULD NOT BE MIXED ON OTHER PLAОВR LINES.

(10-58)

ENTER ALL PLATES USED.

(63-68)

ENTER PLATE SURFACE AREA FACTOR TO BE USED IN ANODE
CALCULATIONS. THIS FACTOR SHOULD BE 1.0 FOR ALL PLATES EXPOSED
TO SEAWATER ONLY ON ONE SIDE AND SHOULD BE 2.0 FOR ALL PLATES
EXPOSED ON BOTH SIDES.

(70-75)

ENTER COST OVERRIDE.

[illegible]

WEIGHT CONTROL OPTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY WEIGHT CONTROL EXECUTION. IT IS USED TO SPECIFY THE OVERALL ANALYSIS PARAMETERS.
(8-13)	SELECT REPORTS.
(8- 9)	ENTER 'PT' IF THE DISCIPLINE DESCRIPTIONS ARE TO BE LISTED.
(10-11)	ENTER 'PT' IF THE ITEM DESCRIPTIONS ARE TO BE LISTED.
(12-13)	ENTER 'PT' IF EACH ITEM OF EACH DISCIPLINE CONTAINED IN THE CONFIGURATION IS TO BE DETAILED.
(21-22)	ENTER THE DISCIPLINE CODE FOR STRUCTURAL ELEMENTS IN THE MODEL INPUT FILE.
(23-24)	ENTER THE DISCIPLINE CODE FOR LOAD LINES IN THE MODEL INPUT FILE THAT ARE NOT ASSIGNED TO A USER-DEFINED DISCIPLINE (NO DISCIPLINE CODE IN COLUMNS 73-74).
(31-32)	LEAVE BLANK IF DISCIPLINE DESCRIPTIONS FOLLOW IN THIS INPUT FILE. ENTER 'XF' IF DISCIPLINES ARE DESCRIBED IN A LIBRARY FILE CONTAINING ONLY DISCIPLINE DESCRIPTIONS.
(33-34)	LEAVE BLANK IF ITEM DESCRIPTIONS FOLLOW IN THIS INPUT FILE. ENTER 'XF' IF ITEMS ARE DESCRIBED IN A LIBRARY FILE CONTAINING ONLY ITEM DESCRIPTIONS.
(35-36)	ENTER 'XF' IF DISCIPLINE AND ITEM DESCRIPTIONS ARE BOTH IN THE SAME EXTERNAL LIBRARY FILE. BECAUSE THIS OPTION IS USED IN LIEU OF THE PRIOR TWO OPTIONS, THE DISCIPLINE AND ITEM EXTERNAL FILE FIELDS SHOULD BE LEFT BLANK WHEN USING THIS OPTION.

LINE LABEL	REPORT SELECTIONS			STRUCTURE DISCIPLINE CODE	MISC. LOADS DISCIPLINE CODE	AUXILIARY FILES			LEAVE BLANK
	DISCIPLINE DESCRIPTION	ITEM DESCRIPTION	DETAIL ITEMS FOR EACH DISCIPLINE			DISCIPLINE DESCRIPTION	ITEM DESCRIPTION	DISCIPLINE AND ITEM DESCRIPTION	
WTCTL									
1-- 5	8-- 9	10--11	12--13	21--22	23--24	31--32	33--34	35--36	37--80
DEFAULT				'ST'	'MI'				